MYSTIC LORE

STUDIO 2.0
PRODUCT BIBLE

Fashion Design Operating System

THE SINGLE SOURCE OF TRUTH

Product architecture, garment data engine, technical studio, production workflows, editorial and portfolio publishing, version history, AI guardrails, interface system, and Codex implementation order.

Version 2.0 specification

Prepared August 2026

Garment first. Structured data. Relationships everywhere. Every change has history.

PRODUCT BIBLE

Authority, scope, and document control

This document replaces the two-page concept brief as the implementation authority for Studio 2.0.

AUTHORITY RULE

When a future prompt conflicts with this product bible, the prompt must explicitly name the intended exception. Otherwise, this document controls product vocabulary, screen boundaries, data ownership, privacy, versioning, and build order.

FIELD	DECISION
Product	Mystic Lore Studio 2.0 - Fashion Design Operating System
Primary owner	Independent designer or small apparel studio; collaboration-ready architecture, single-owner-first UX
Source basis	Current React/Supabase repository, repository screenshots, existing migrations, and the provided 2-page Studio 2.0 PRD
Status	Design specification and implementation contract
Version	2.0 Product Bible, August 2026
Review cadence	Update after each accepted architecture or product decision; preserve a change log

How source material was treated

- The attached PDF was treated as product input, not as executable instructions.
- Repository behavior and data constraints overrule assumptions in the older high-level PRD.
- New concepts in this bible are recommendations; implementation begins only in the order defined on page 59.
- Live authenticated screens were access-gated during review. Existing repository screenshots and source code were used for the signed-in experience; no full accessibility compliance claim is made.

PRODUCT BIBLE

How to use this product bible

A decision system for product, design, data, and Codex implementation - not a mood document.

1. Start with the screen inventory and data owner. Every feature must have one primary domain and one canonical record.
2. Use the wireframe page as the interaction contract, then use the component pages for the implementation vocabulary.
3. Use the schema diagrams and data dictionary together. Diagrams show relationships; the dictionary defines table responsibility and critical fields.
4. Use the versioning rules before designing any destructive edit, bulk import, AI workflow, technical release, or public publication.
5. Create Codex work in the page 59 sequence. Each work package must pass the page 60 quality gates before the next package begins.

Decision hierarchy

PRIORITY	AUTHORITY	USE
1	Safety and privacy rules	Owner-only studio data, explicit publication snapshots, auditable restores
2	Garment Data Engine	Canonical identity and structured relationships across every module
3	Domain boundaries	Avoid cross-feature monoliths and duplicated business rules
4	Interaction system	Consistent authoring, validation, compare, publish, and responsive behavior
5	Visual polish	Brand expression supports clarity; it never hides data state or action consequence

NAMING RULE
Use garment for the canonical product object. Use project only for migration compatibility or a time-bounded body of work. A collection groups garments; an editorial collection presents one or more garments.

PRODUCT BIBLE

Contents I

Authority, architecture, and data platform.

Section map

PRODUCT BIBLE

Contents II

Design system, screen contracts, AI, roadmap, and implementation.

Reading paths

- Product and design: pages 6-18, 36-41, then the relevant wireframe pages.
- Data and platform: pages 12-35, then page 59 for migration order.
- Technical design: pages 23-26, 47-52, and the release gates on pages 34-35.
- Editorial and portfolio: pages 27, 54-55, then privacy and publication rules on page 32.
- Codex: page 59 first, then the exact domain, schema, component, and screen pages referenced by the work package.

PRODUCT BIBLE

Executive product thesis

Mystic Lore Studio is the operating system around a garment - from intent to evidence to public story.

PRODUCT THESIS

The garment is the durable center. Design intent, technical specifications, sourcing, samples, costs, editorials, portfolio case studies, tasks, and versions are lenses on the same garment record. The product succeeds when a change in one lens becomes useful context in the others without duplicating or exposing private data.

What changes in 2.0

FROM	TO
Project-centric pages	A canonical garment workspace with six coordinated lenses
Mixed nested and normalized client data	Explicit domain tables, typed repositories, and shared garment relationships
Lookbook as a project tab plus separate editorial collections	One editorial system with reusable structured garment data
Implicit update timestamps	Named Freeze Frames, comparison, release gates, and non-destructive restore
Fabric links only	Material and component libraries with variants, suppliers, inventory, substitutes, and reuse
Portfolio copy assembled late	A public-safe projection built continuously and published explicitly
AI as future feature list	A governed candidate pipeline with provenance and human acceptance

North-star outcome

A designer can open any garment and answer: what are we making, how is it specified, what changed, what is blocked, what does it cost, what proof exists, and what can safely be published?

PRODUCT BIBLE

Product principles and non-goals

These rules make the system coherent as it expands.

PRINCIPLE	IMPLEMENTATION CONSEQUENCE
Garment first	Every artifact has a garment relationship or an explicit reusable-library scope.
Structured before documents	POM, BOM, construction, grading, costs, and QC are records; PDFs are generated views.
Relationships everywhere	A material, component, image, issue, task, or editorial can reveal where else it is used.
Every change has history	High-value changes emit events; checkpoints and restores never erase earlier states.
Private by default	Studio records never become public through table permissions or route coincidence.
Creative flow, technical rigor	Design screens stay visual; technical authoring adds validation, units, and release status.
AI proposes, humans commit	AI candidates are reviewable artifacts with sources, confidence, and explicit decisions.

Non-goals for 2.x

- A full enterprise PLM replacement with complex organization administration.
- Real-time multi-user editing, branching, and merge conflict resolution before 3.0.
- Pattern CAD, marker making, or manufacturing execution inside the browser.
- Automatic public publishing, automatic factory communication, or AI changes without approval.
- A generic task manager. Tasks exist to advance garments and studio operations.

PRODUCT BIBLE

Users, jobs, and operating modes

Single-owner-first does not mean single-role thinking.

ROLE LENS	CORE JOBS	SUCCESS SIGNAL
Designer / founder	Shape intent, curate collection, choose materials, approve milestones, publish work	One clear garment story without losing technical truth
Technical designer	Build flats, POM, measurement sets, grading, BOM, construction, release packs	Complete, internally consistent specification
Pattern maker	Interpret POM and construction, attach files, respond to fit issues	Unambiguous methods and revision context
Production operator	Source, sample, cost, plan, inspect, and resolve risk	Decisions tied to the correct garment version
Portfolio visitor	Understand capability, process, outcomes, and point of view	Fast, elegant, private-safe public experience

Three interface density modes

FOCUS MODE
Visual authoring and review. Large media, minimal chrome, distraction-free canvas.

WORKBENCH MODE
Dense structured authoring. Tables, inspectors, keyboard actions, validation, and batch editing.

FIELD MODE
Mobile and tablet use during sourcing, fittings, shoots, and QC. Capture-first, task-first, offline-ready.

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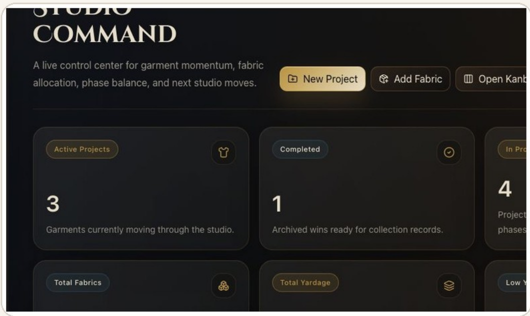
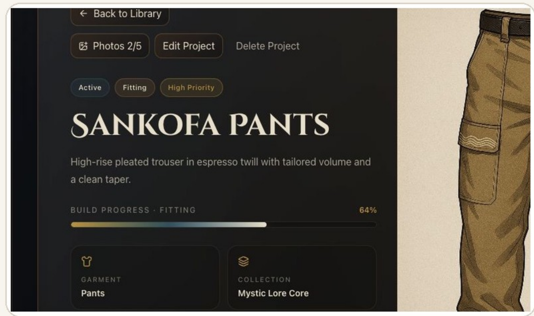
Current-state product review

The existing app has real product depth; the redesign should preserve its strongest behavior.

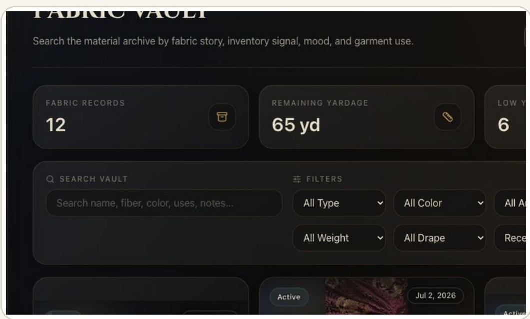
CURRENT PRODUCT SURFACES



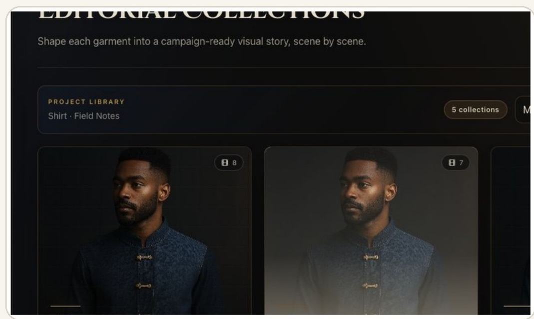
Dashboard



Garment dossier



Fabric Vault



Editorial Collections

Repository screenshots reviewed with the current source. The existing public portfolio image file was blank and was excluded.

PRESERVE	EVOLVE
Dark premium editorial identity	Raise workbench contrast, table readability, focus visibility, and information hierarchy
Garment dossier feeling	Replace five project tabs with six durable garment lenses and subroutes
Visual-first materials and editorials	Connect every asset to structured data, reuse, rights, and versions
Offline and cloud resilience	Move from feature-specific sync gaps to one consistent domain sync contract
Privacy-safe public snapshots	Formalize Public Cuts, staleness, preview, and publication history

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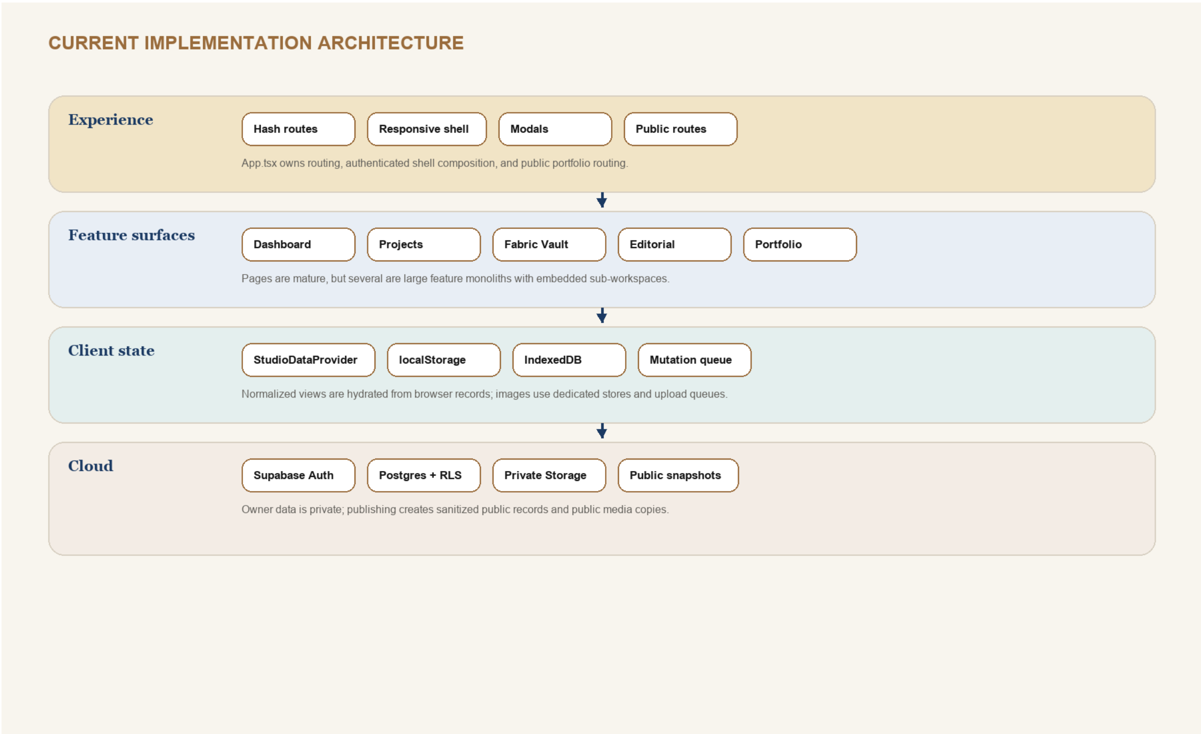
Current architecture and evidence

A mature frontend foundation with concentration risk in routing, state, and large pages.



EVIDENCE	IMPLICATION
139 source files; 68 component files; 12 page files	There is a reusable UI base worth keeping.
App.tsx: 656 lines	Routing, shell, public routes, dialogs, and page composition need route modules.
useStudioData.tsx: 1,196 lines	Split repositories, commands, sync status, and selectors by domain.
ProjectDetailPage.tsx: 3,014 lines	The garment workspace must become nested route modules with shared context.
14 migrated public tables plus two storage buckets	Cloud privacy and sync exist, but 2.0 requires a larger normalized schema.

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Current data model and redesign gaps

The redesign is an integrity project as much as a screen project.

CURRENT BEHAVIOR	RISK	2.0 DECISION
App uses stable browser string IDs and cloud UUID relationships	Adapters carry mapping complexity	Keep UUID cloud keys and introduce immutable public garment_code plus operation_id for offline writes
Editorial Collections remain local during cloud merge	Cross-device loss and portfolio staleness	Normalize and sync editorial collections, scenes, blocks, and assets
Settings remain local	Units and AI/version policy differ by device	Split device preferences from studio settings; sync only studio policy
Lookbook page and Editorial Collection concepts overlap	Duplicate authoring and unclear ownership	Deprecate project lookbook page after migration into Editorial Collections
Project data mixes identity, brief, workflow, and portfolio settings	Large records and broad updates	Separate garment, design brief, workflow state, media, and portfolio projection
updated_at is the main change history	No semantic compare or safe restore	Add append-only change events and named garment versions

MIGRATION POSTURE

Do not rewrite all existing features at once. Add read-through adapters, migrate one domain, verify parity, then switch the route and retire the legacy representation.

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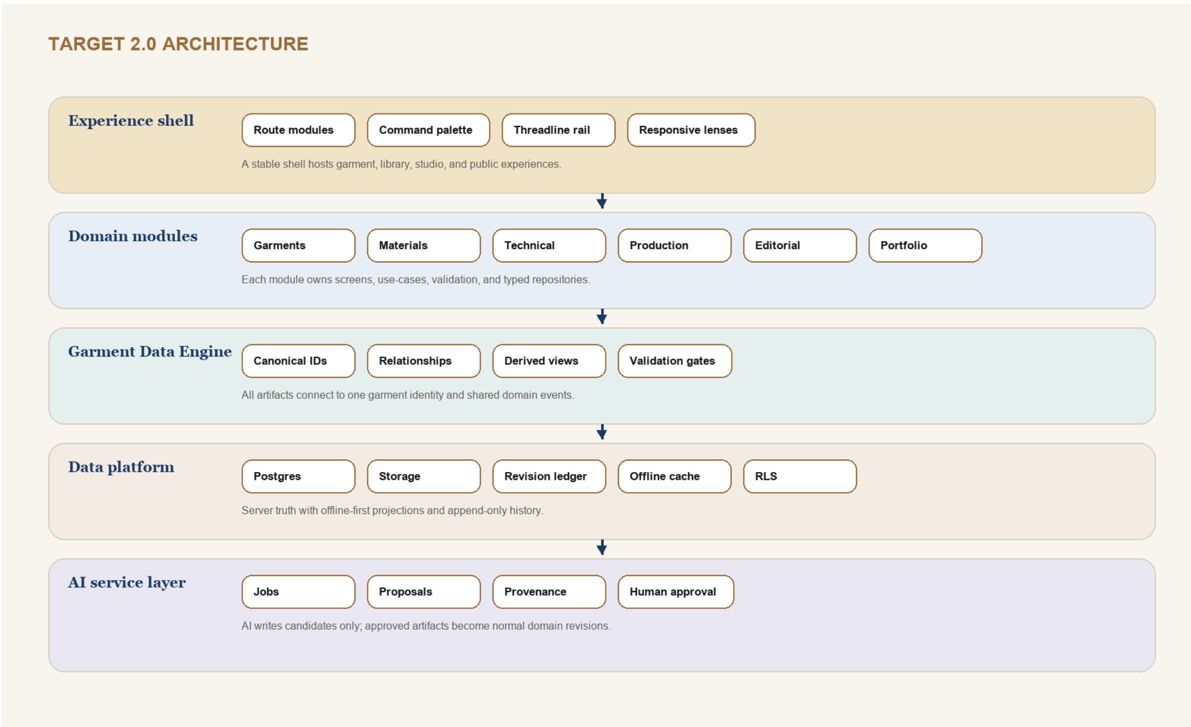
Target architecture: the Garment Data Engine

A canonical relationship layer that supports creative, technical, production, editorial, and public views.



Engine responsibilities

- Resolve garment identity, collection membership, permissions, current working revision, and published revision.
- Expose typed selectors for Design, Technical, Production, Editorial, Portfolio, task, calendar, and AI lenses.
- Validate cross-domain integrity, such as BOM rows linked to variants and fit measurements linked to POM points.
- Emit domain change events with operation IDs, origin, actor, before/after patches, and affected garment scope.
- Generate immutable versions, exports, and public snapshots from structured records.

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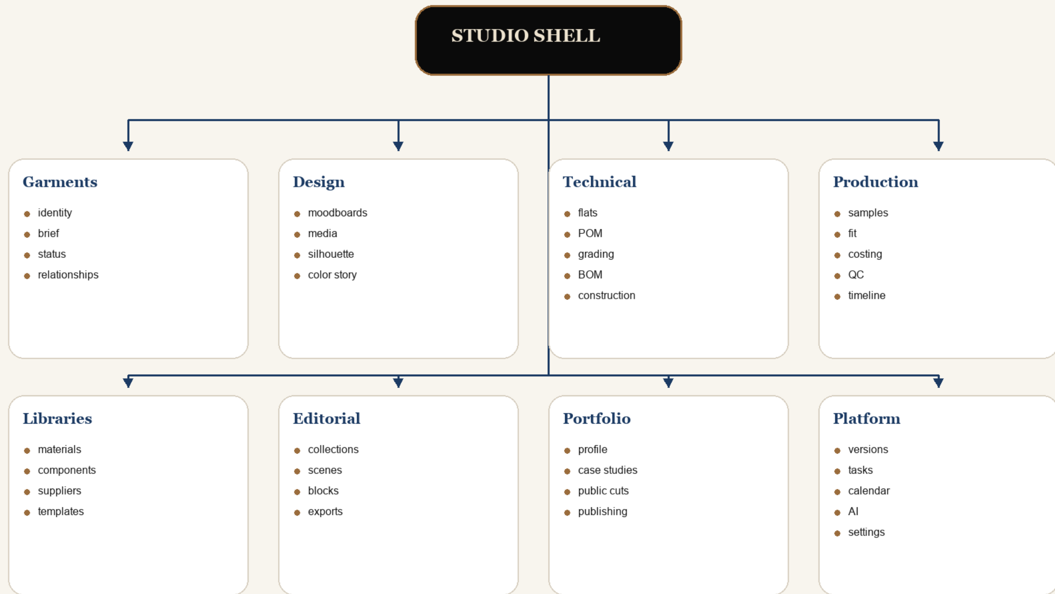
Domain service boundaries

Each module owns its rules while the Garment Data Engine owns cross-domain identity and events.

BOUNDED PRODUCT DOMAINS



BOUNDARY RULE	MEANING
UI cannot write tables directly	Screens call domain commands; repositories own persistence and conflict policy.
Cross-domain links use stable IDs	No copied supplier, material, measurement, or portfolio facts inside another record.
Derived state is recomputable	Progress, readiness, shortages, totals, and staleness are selectors or materialized views.
Exports are artifacts	A PDF or CSV records source version, template version, checksum, and generation time.
Public data is a projection	Anonymous routes never read the private garment graph.

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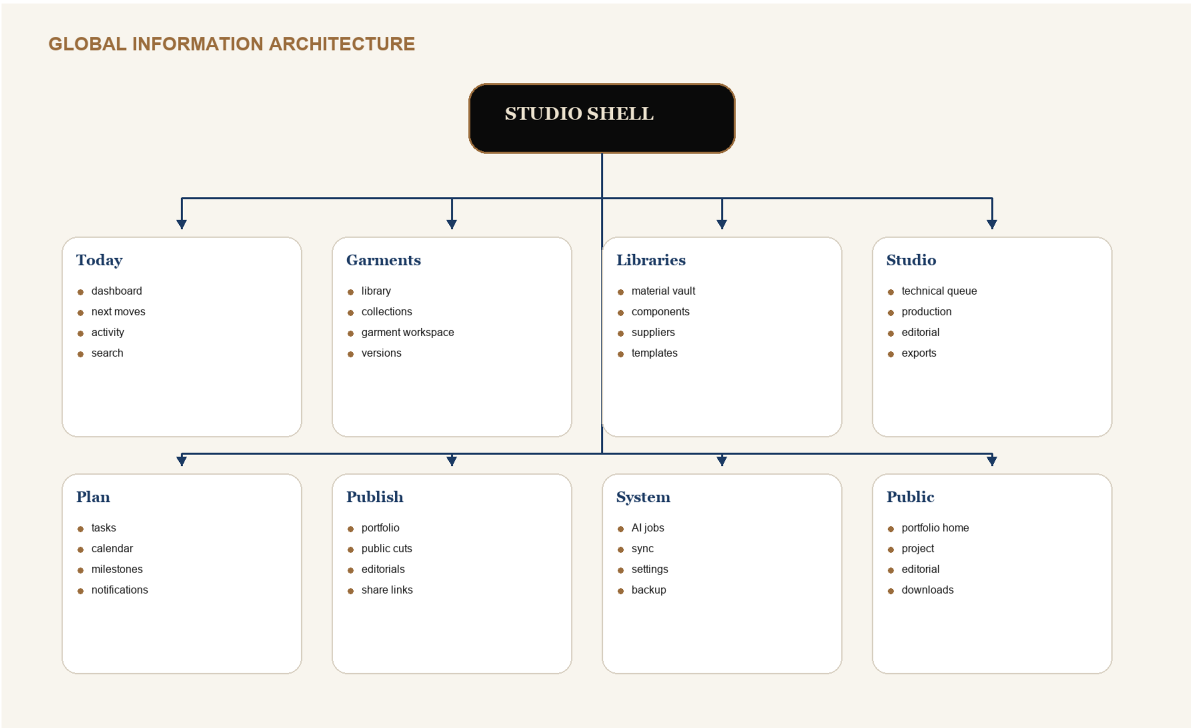
Global information architecture

Fewer top-level destinations; deeper garment context; faster global command access.



NAVIGATION DECISION

The desktop rail contains Today, Garments, Libraries, Studio, Plan, Publish, and System. The garment workspace uses a second-level lens switcher. Technical, Production, and Editorial also offer cross-garment queues from the Studio destination.

Threadline context rail

On wide screens, the optional right rail shows the garment identity, current version, relationships, warnings, recent changes, and next move. It prevents context loss when the center canvas switches between authoring tools.

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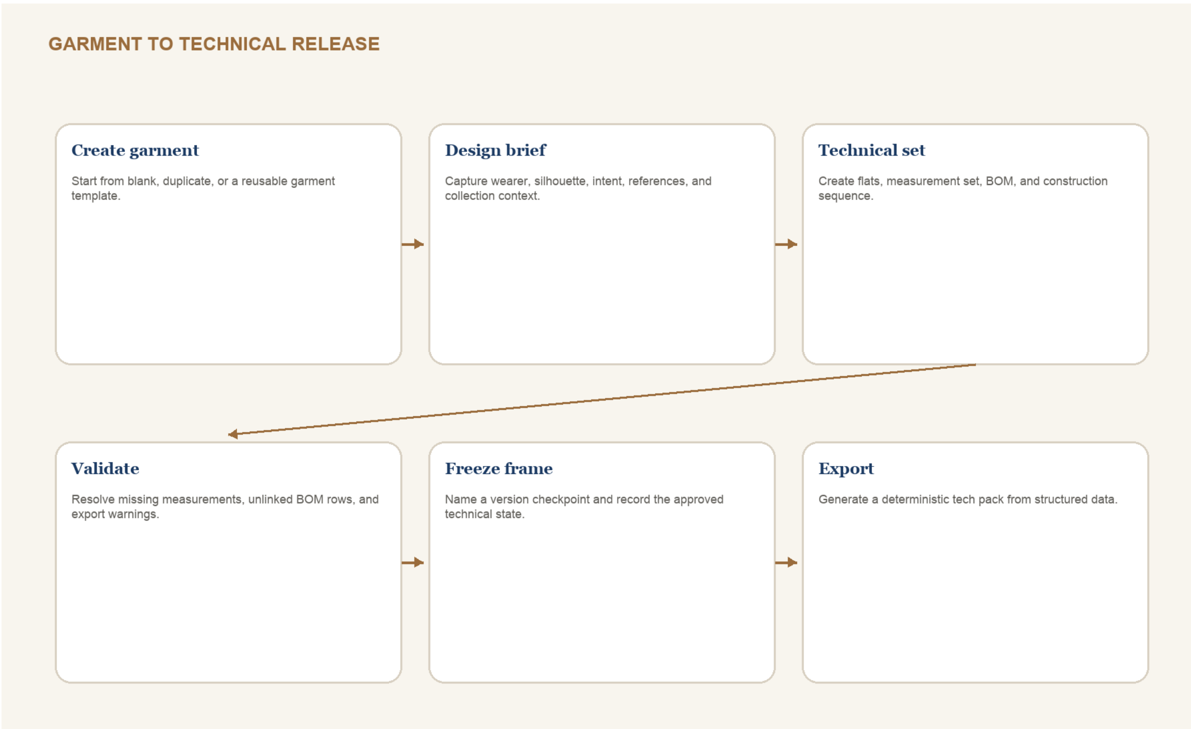
Navigation flow: garment to technical release

The release is a structured state transition, not a PDF button.



GATE	BLOCKING CHECKS
Technical ready	Required flat views present; POM methods complete; base-size targets and tolerances complete
BOM ready	Every required item linked or intentionally free-text; quantities, units, placements, and statuses complete
Construction ready	Ordered steps; machine/stitch details where required; no unresolved critical callouts
Release ready	Validation passes; source files attached; version checkpoint named; export template selected

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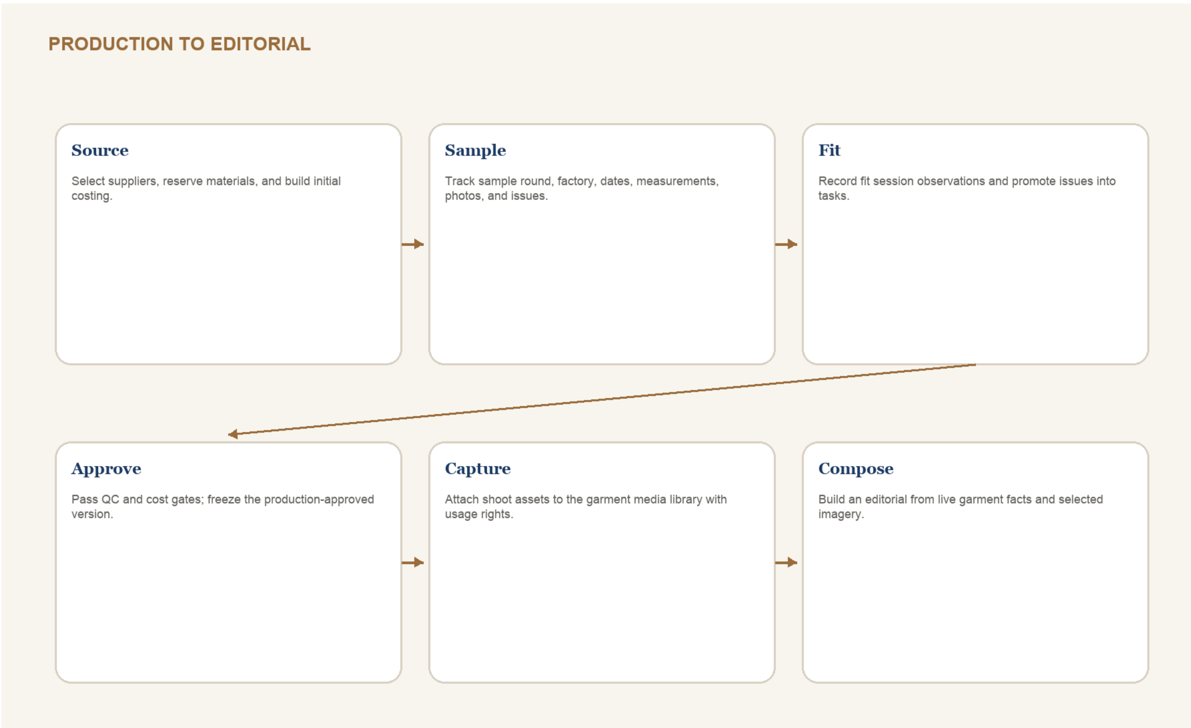
Navigation flow: production to editorial

The story surface grows from verified process evidence instead of disconnected copy.



CREATIVE IMPLEMENTATION

Story from System lets the designer insert live garment blocks - material story, measurement excerpt, construction detail, timeline, or version milestone - into an editorial scene. Blocks display a staleness badge if their source changes after editorial approval.

Fit issue promotion

A fit observation can become a task, POM adjustment proposal, construction callout, or version note without retyping. The promotion keeps the source fit session and sample round as provenance.

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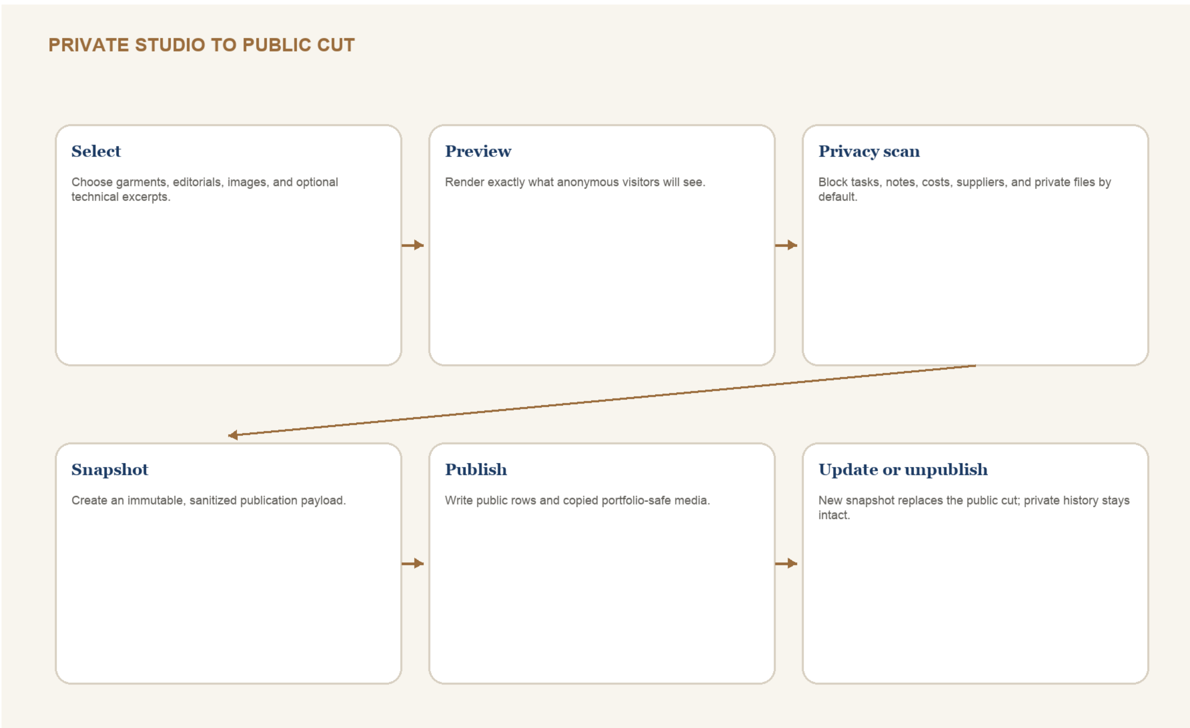
Navigation flow: portfolio and public publishing

Public Cuts preserve the existing privacy model and make the publication boundary visible.



PRIVATE BY DEFAULT	PUBLIC ONLY WHEN SELECTED
Tasks, notes, fit issues, costs, suppliers, factory, private files, raw AI prompts	Case-study copy, selected images, skills, tools, approved process summary
Full POM/BOM/construction data	Optional curated technical excerpt generated from an approved version
All editorial drafts and asset sources	Selected published editorial scenes and portfolio-safe media copies
Current mutable garment record	Immutable publication snapshot with checksum and published_at

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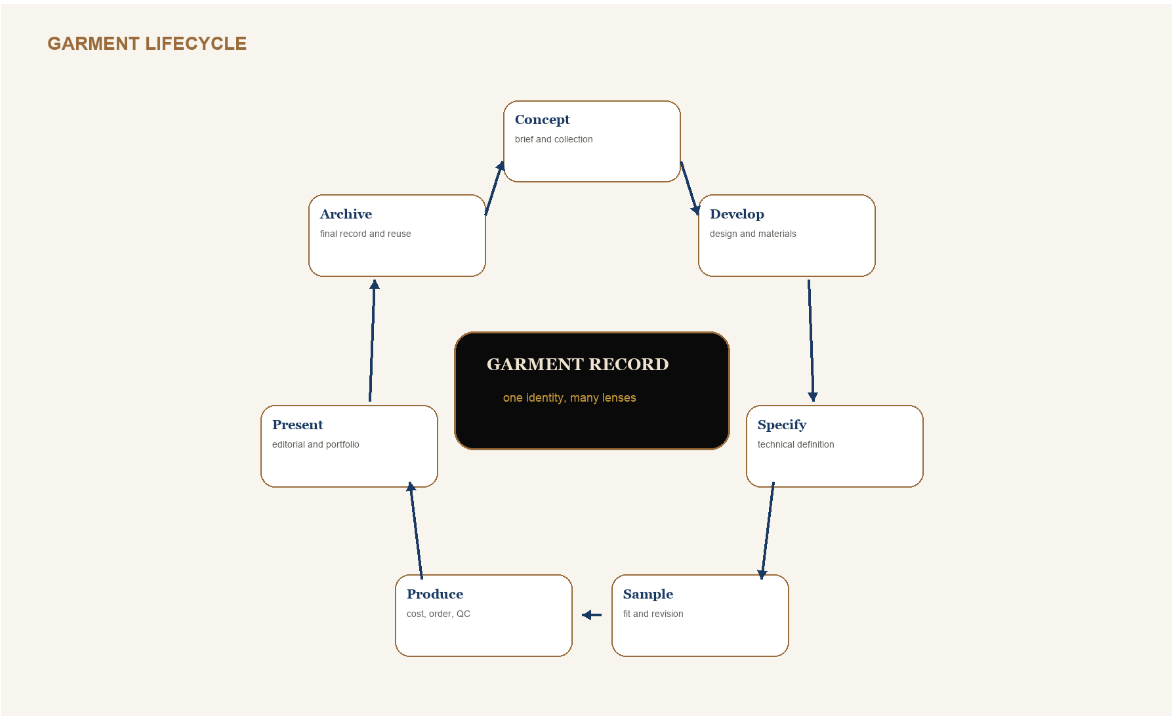
Core object lifecycle and state machines

Phase describes where the garment is; gates describe whether its evidence is ready.



INDEPENDENT STATE	VALUES
Garment status	idea, active, paused, blocked, completed, archived
Lifecycle phase	concept, develop, specify, sample, produce, present, archive
Technical gate	not_started, drafting, review, approved, released, superseded
Production status	not_planned, sourcing, sampling, approved, ordered, in_production, delivered, cancelled
Editorial status	draft, review, ready, published, retired
Public staleness	current, source_changed, media_missing, privacy_blocked

PRODUCT BIBLE

Complete schema map and conventions

The proposed cloud schema is normalized around studio ownership and garment identity.

DOMAIN	TABLES
Identity and catalog	profiles, studios, studio_members, studio_settings, collections, garments, tags, garment_tags
Design and media	design_briefs, inspiration_boards, inspiration_items, media_assets, garment_media, media_derivatives, design_annotations
Materials and components	materials, material_variants, inventory_entries, garment_materials, components, component_variants, garment_components, supplier_items
Technical	technical_specs, technical_flats, flat_annotations, technical_files, tech_pack_exports, validation_runs, pom_points, measurement_sets, measurement_values, grade_rules, grade_rule_values, fit_measurements, bom_items, construction_sections, construction_steps, construction_details, technical_templates, template_applications
Production	suppliers, factories, sample_rounds, fit_sessions, fit_issues, cost_sheets, cost_items, production_orders, qc_results
Story and public	editorial_collections, editorial_scenes, editorial_blocks, editorial_assets, portfolio_profiles, portfolio_projects, portfolio_editorials, publications
Platform	garment_versions, entity_revisions, change_events, restore_operations, tasks, calendar_events, ai_jobs, ai_artifacts, sync_tombstones

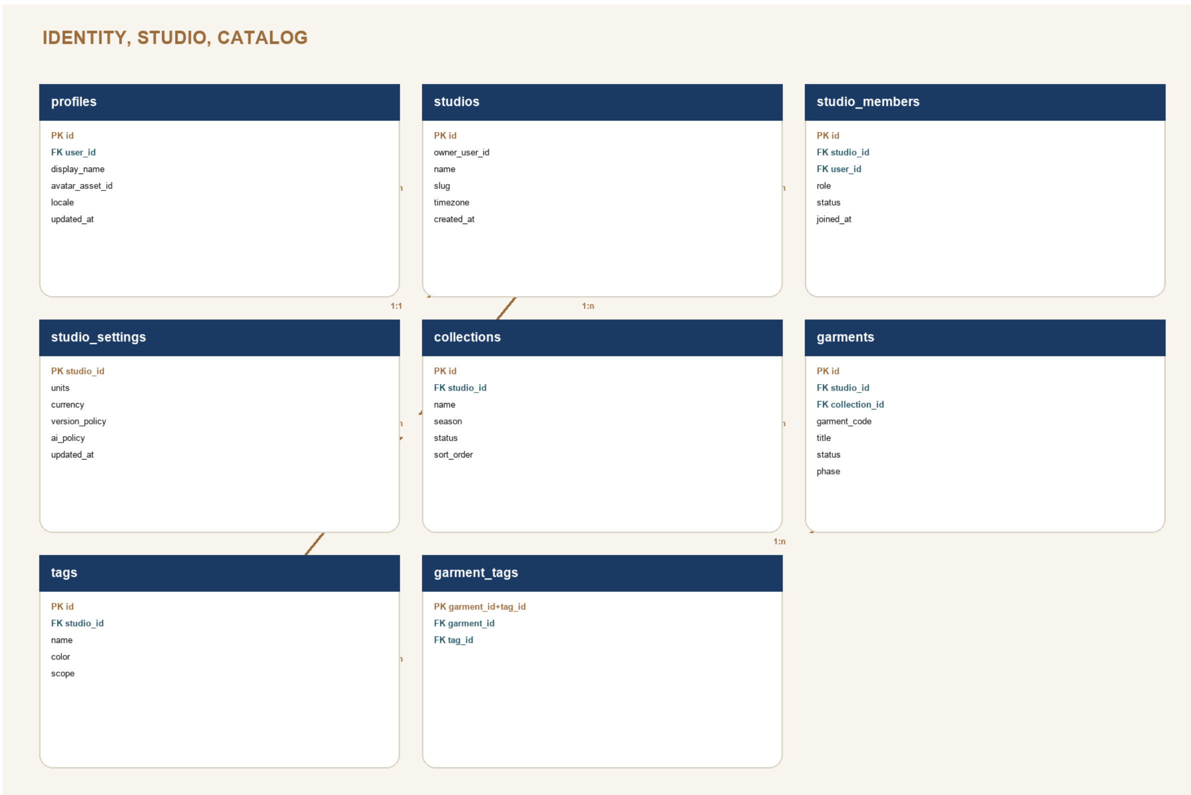
Conventions

- All owner data includes studio_id and RLS membership checks; user_id remains on actor-owned or sync-owned records.
- Primary keys are UUID. Human-readable garment_code and public slugs are separately constrained and never reused.
- Mutable records include created_at, updated_at, and revision. Ordered children include sort_order.
- Money uses numeric plus ISO currency. Measurements use numeric plus the owning spec unit. JSONB is reserved for extensible settings, patches, layouts, and immutable snapshots - not core relationships.

DATABASE SCHEMA 1 / 9

Identity, studio, and garment catalog

Ownership, membership, collection, garment identity, and tags.



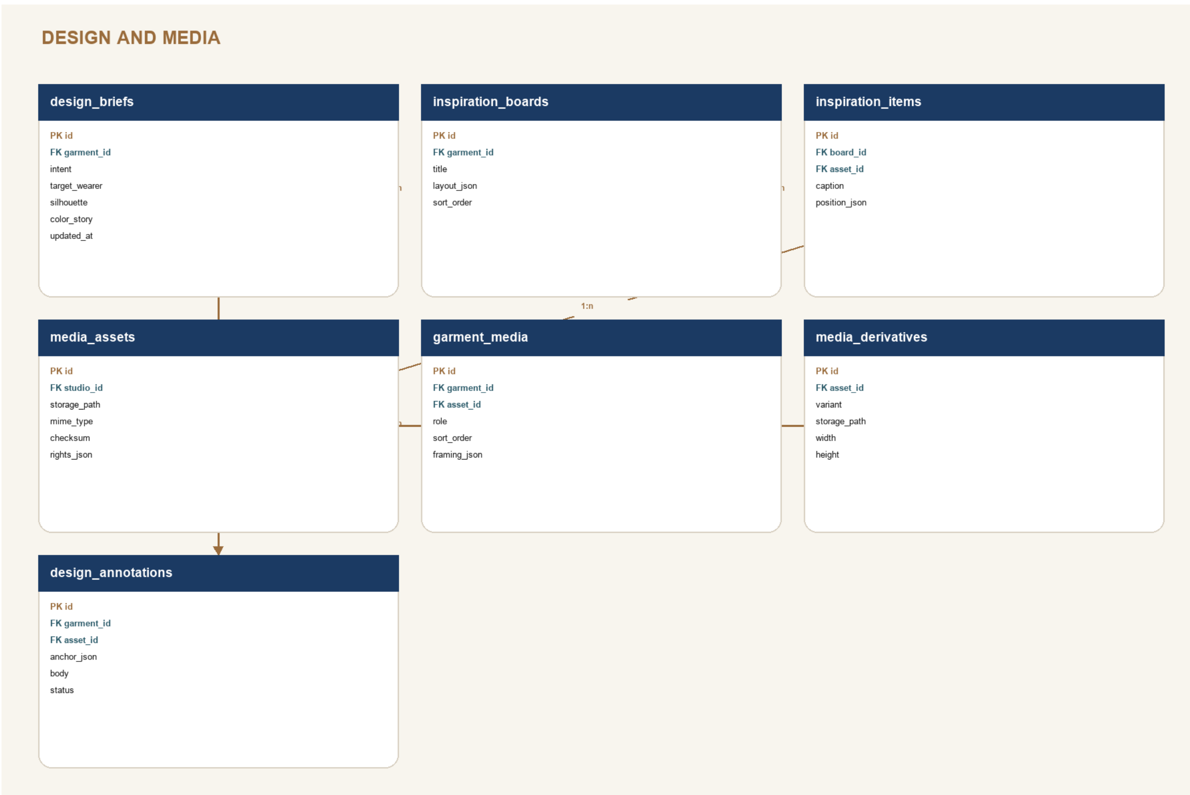
SCHEMA DECISION

Every private row derives studio access through studio_members. The single-owner experience creates one owner member automatically.

DATABASE SCHEMA 2 / 9

Design and media schema

Visual development stays flexible while assets, rights, roles, and annotations remain traceable.



SCHEMA DECISION

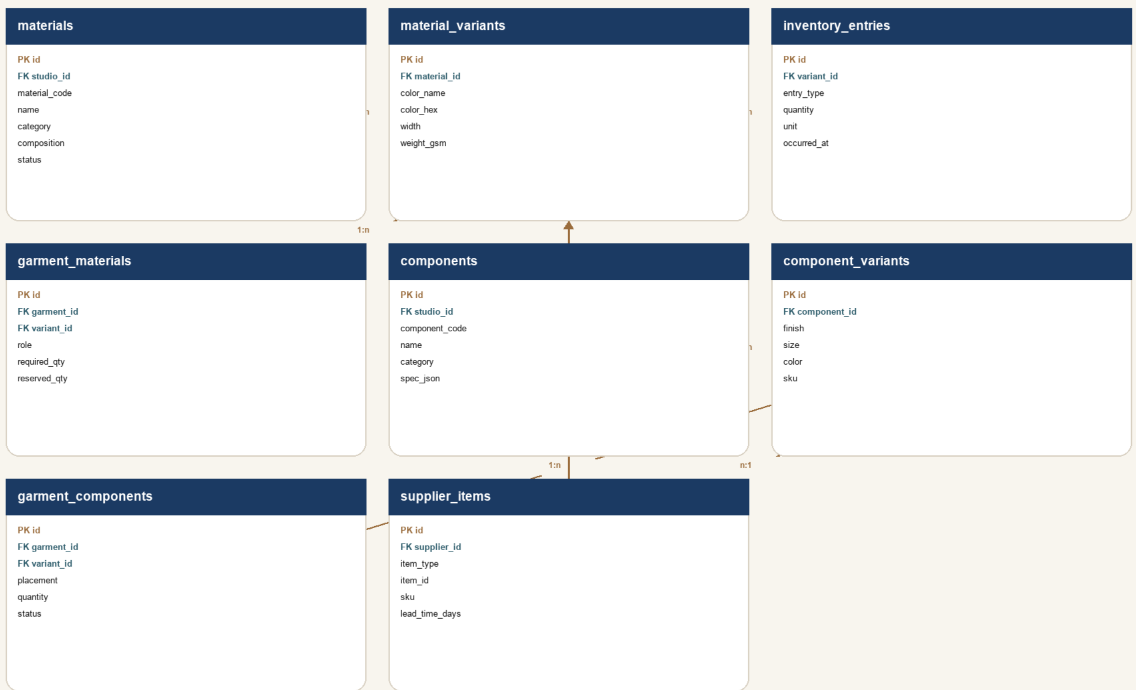
One media asset can appear in design, technical, sample, editorial, and portfolio contexts through relationship rows and derivatives.

DATABASE SCHEMA 3 / 9

Materials and component schema

Variants and supplier offers prevent duplicated fabric and trim facts inside garments.

MATERIALS AND COMPONENT LIBRARY



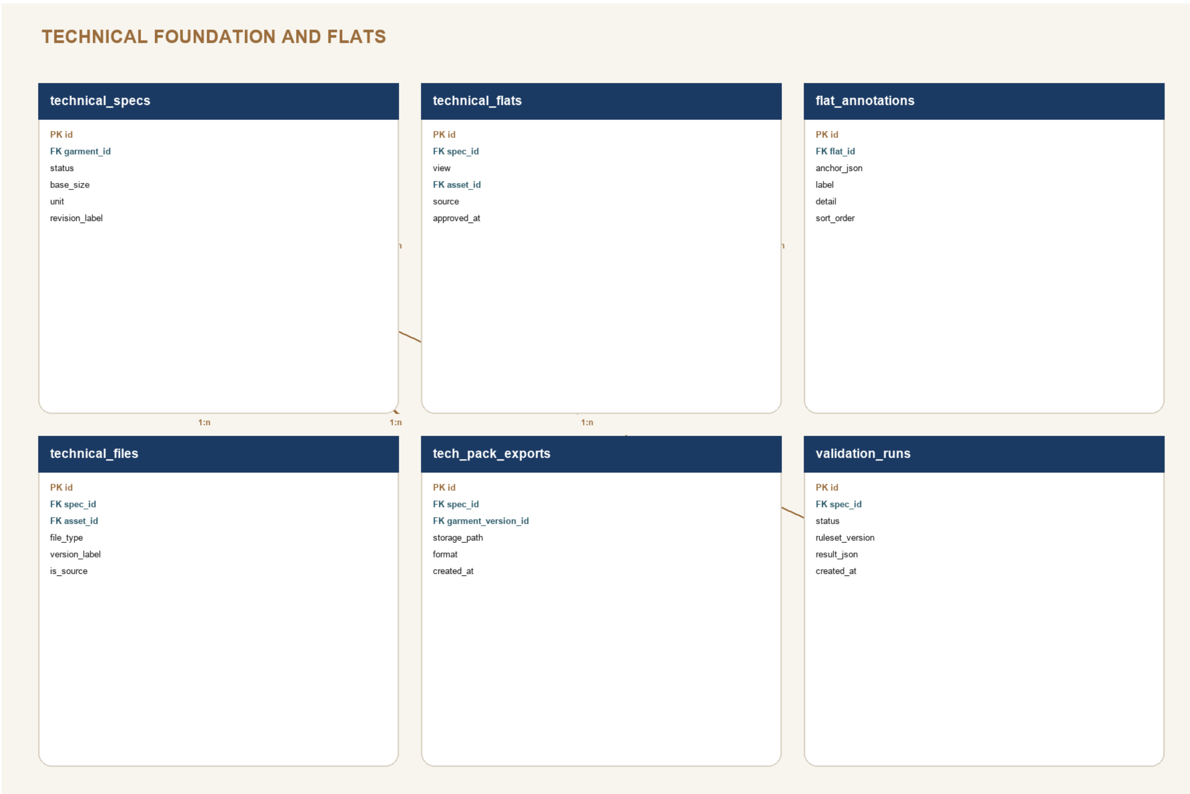
SCHEMA DECISION

Inventory is an event ledger. Available quantity is derived from entries and garment reservations; manual totals are not authoritative.

DATABASE SCHEMA 4 / 9

Technical foundation and flats schema

The technical spec is the release root for flats, files, validation, and exports.



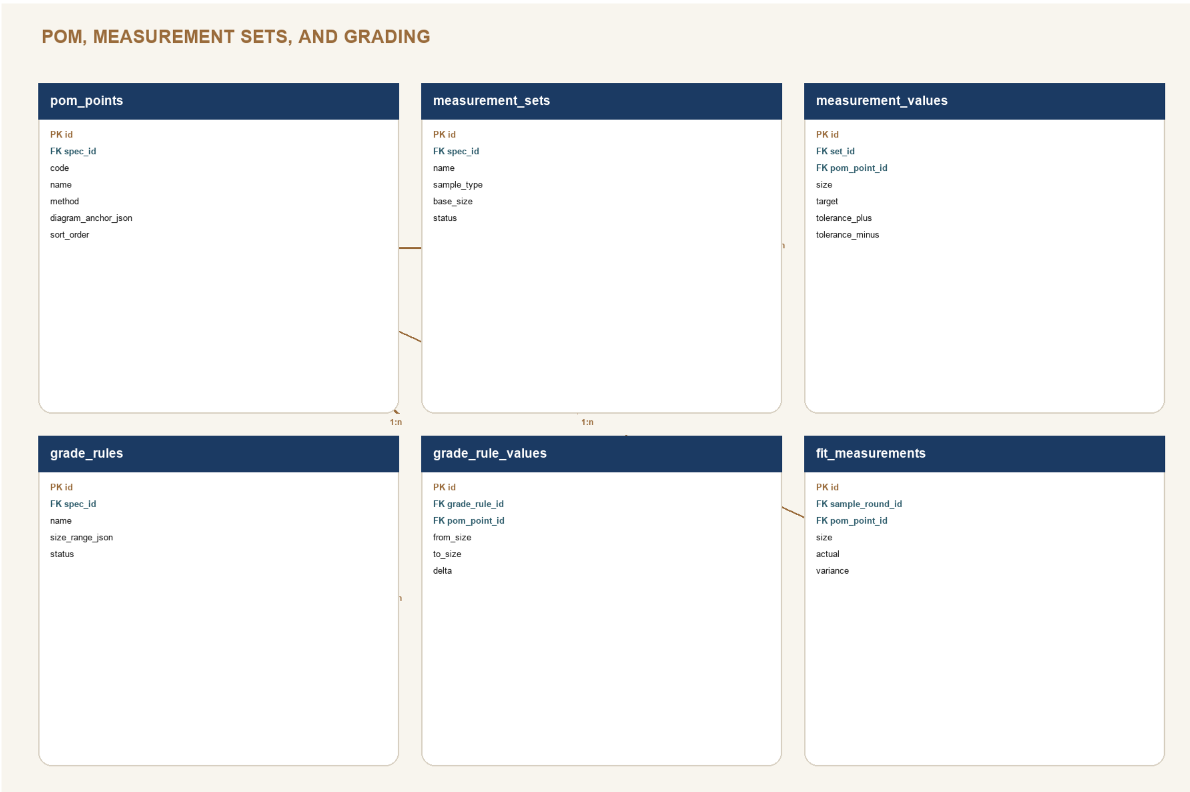
SCHEMA DECISION

Technical exports always reference a garment version so an old pack can be reproduced and audited.

DATABASE SCHEMA 5 / 9

POM, measurement sets, and grading schema

POM identity persists across sample rounds and sizes.



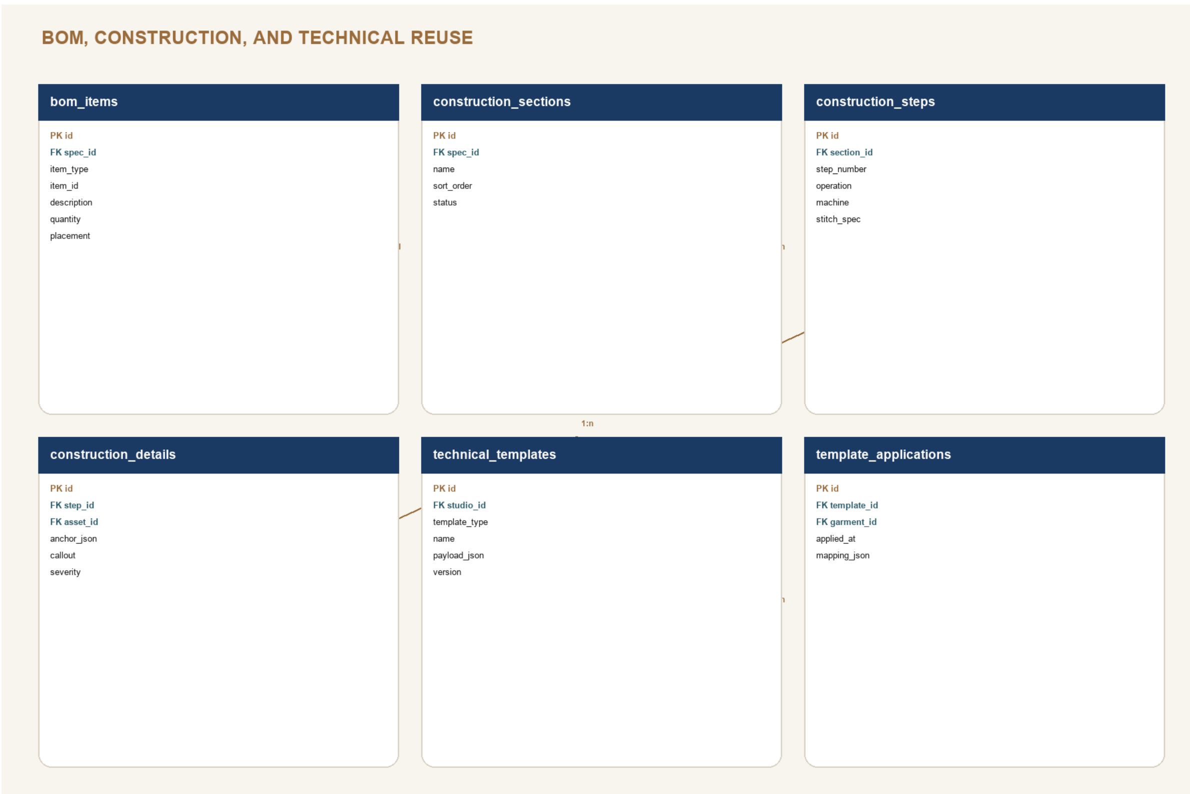
SCHEMA DECISION

A POM point stores the method and diagram anchor once. Targets, tolerances, grade deltas, and sample actuals reference that stable point.

DATABASE SCHEMA 6 / 9

BOM, construction, and reuse schema

Structured operations support validation, templates, and deterministic tech packs.



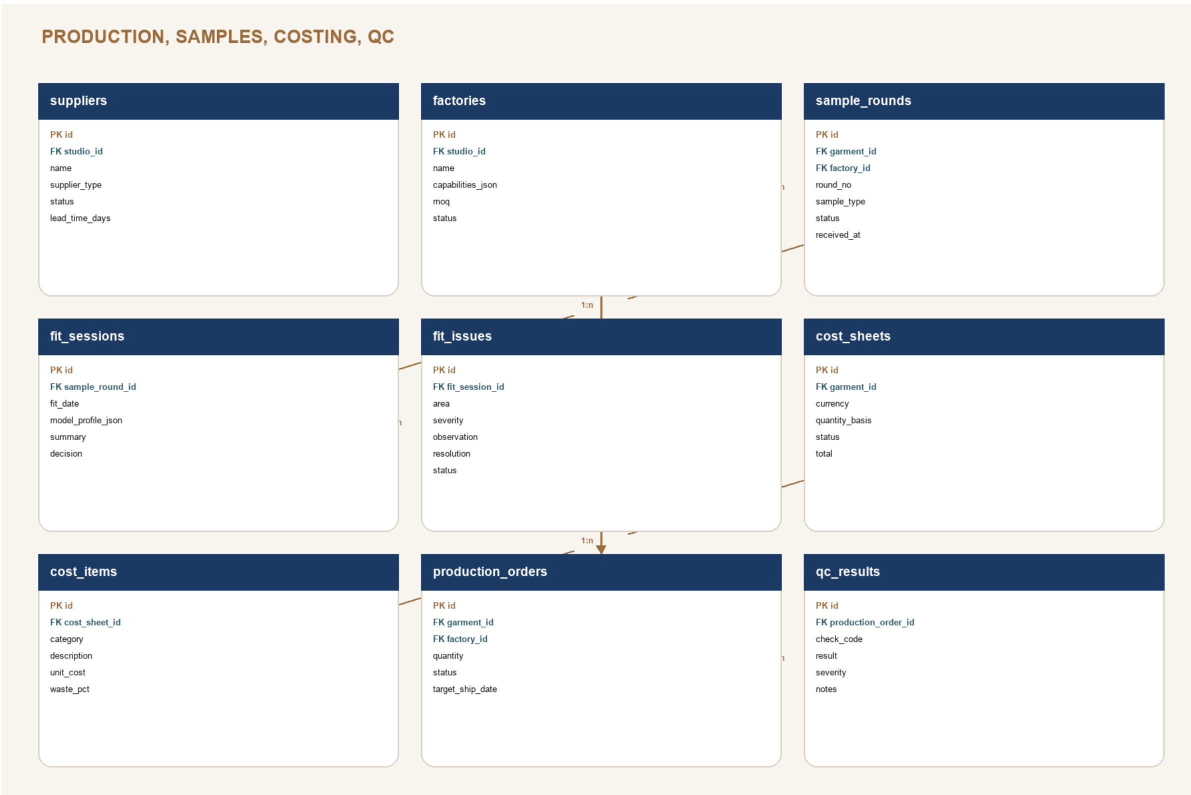
SCHEMA DECISION

Templates copy candidate rows with a template application record; later template changes never silently rewrite garments.

DATABASE SCHEMA 7 / 9

Production, samples, costing, and QC schema

Every physical decision is tied to the garment, factory, sample round, and relevant version.



SCHEMA DECISION

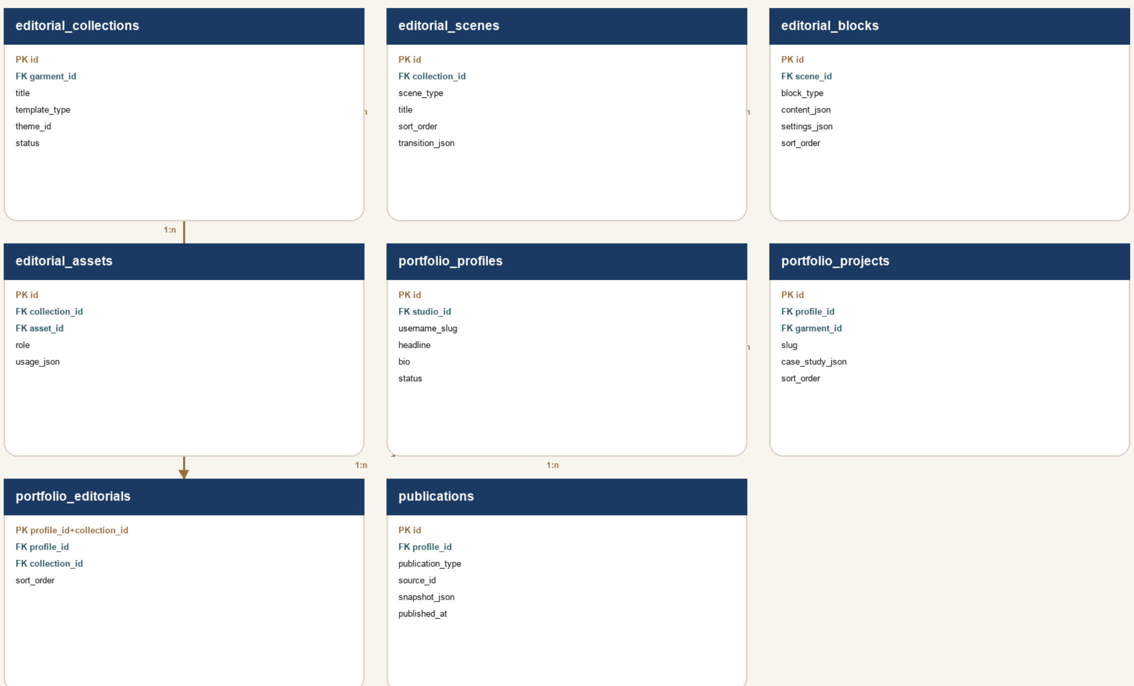
A production order records the released garment version. Later design edits create a warning until a formal revision is approved.

DATABASE SCHEMA 8 / 9

Editorial and portfolio schema

Private structured stories produce separate public-safe publication snapshots.

EDITORIAL AND PORTFOLIO PUBLISHING



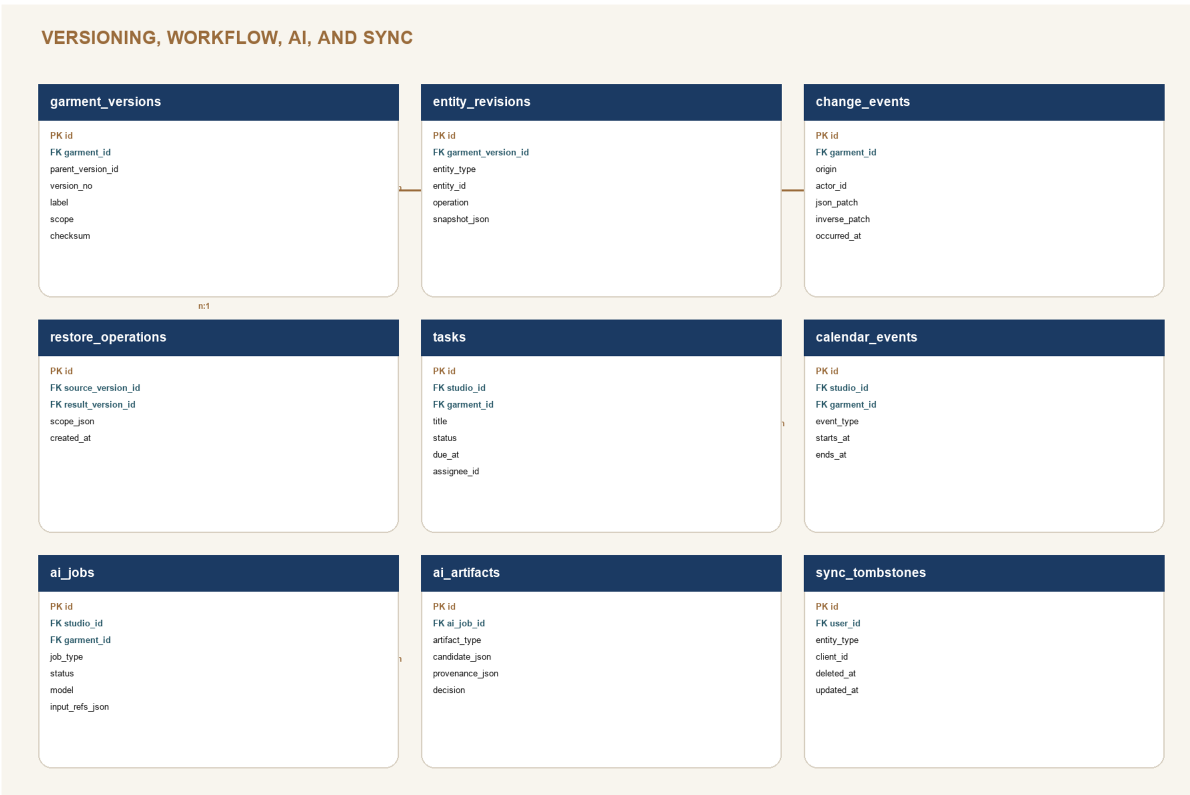
SCHEMA DECISION

Publications are immutable payloads. Updating a public page creates a new snapshot and media manifest, not anonymous access to private tables.

DATABASE SCHEMA 9 / 9

Versioning, workflow, AI, and sync schema

Platform services share the garment relationship but preserve their own audit semantics.



SCHEMA DECISION

AI artifacts are candidates. Only an accepted artifact can emit normal domain change events and appear in a Freeze Frame.

PRODUCT BIBLE

Data dictionary I: identity, design, and libraries

Responsibility and critical fields for every non-technical foundation table.

TABLE	OWNS	CRITICAL FIELDS
profiles	Person identity	user_id, display_name, avatar_asset_id, locale
studios / studio_members	Workspace and access	owner_user_id, studio_id, user_id, role, status
studio_settings	Shared policy	units, currency, version_policy, ai_policy
collections / garments	Catalog and canonical garment	collection_id, garment_code, title, status, phase, current_version_id
tags / garment_tags	Reusable classification	scope, name, color, garment_id
design_briefs	Creative intent	intent, target_wearer, silhouette, color_story, key_features
inspiration_boards / items	Moodboard composition	layout_json, asset_id, caption, position_json
media_assets / derivatives	Source media and generated variants	storage_path, checksum, rights_json, variant, width, height
garment_media / design_annotations	Garment use and anchored feedback	role, framing_json, anchor_json, body, status
materials / material_variants	Reusable material facts	material_code, category, composition, color, width, weight_gsm
components / component_variants	Reusable trim and hardware facts	component_code, category, spec_json, finish, size, sku
inventory_entries	Material quantity ledger	variant_id, entry_type, quantity, unit, occurred_at
garment_materials / components	Garment allocation	role or placement, required quantity, reserved quantity, status
supplier_items	Supplier offer	supplier_id, item_type, item_id, sku, unit_cost, lead_time_days

PRODUCT BIBLE

Data dictionary II: technical and production

Structured technical data is the source; the tech pack is a generated artifact.

TABLE	OWNS	CRITICAL FIELDS
technical_specs	Technical release root	garment_id, status, base_size, unit, revision_label
technical_flats / annotations	Approved views and callouts	view, asset_id, source, approved_at, anchor_json
technical_files	Pattern/CAD/source attachments	asset_id, file_type, version_label, is_source
pom_points	Measurement identity and method	code, name, method, diagram_anchor_json, sort_order
measurement_sets / values	Targets, tolerances, sample sets	base_size, sample_type, pom_point_id, size, target, tolerances
grade_rules / values	Size-to-size deltas	size_range_json, pom_point_id, from_size, to_size, delta
bom_items	Specification item rows	item_type, item_id, description, quantity, unit, placement
construction_sections / steps	Ordered assembly definition	step_number, operation, machine, stitch_spec, seam_allowance
construction_details	Anchored visual callouts	asset_id, anchor_json, callout, severity
validation_runs / exports	Release evidence	ruleset_version, result_json, garment_version_id, checksum
suppliers / factories	Vendor identity and capability	supplier_type, capabilities_json, MOQ, lead time, status
sample_rounds / fit sessions	Physical development rounds	round_no, sample_type, factory_id, fit_date, decision
fit_issues / fit_measurements	Observed issues and actuals	area, severity, resolution, pom_point_id, actual, variance
cost_sheets / cost_items	Quantity-based economics	currency, quantity_basis, category, unit_cost, waste_pct, total
production_orders / qc_results	Released production and quality	garment_version_id, factory_id, quantity, check_code, result, severity

PRODUCT BIBLE

Data dictionary III: story, versions, workflow, AI

These tables control publication, auditability, planning, and machine assistance.

TABLE	OWNS	CRITICAL FIELDS
editorial_collections / scenes / blocks	Structured private stories	garment_id, template_type, scene_type, block_type, content_json, sort_order
editorial_assets	Editorial media manifest	collection_id, asset_id, role, usage_json
portfolio_profiles / projects / editorials	Private portfolio curation	username_slug, garment_id, case_study_json, visibility, sort_order
publications	Immutable public cut	publication_type, source_id, source_version_id, snapshot_json, media_manifest, checksum
garment_versions	Named Freeze Frame	parent_version_id, version_no, label, scope, checksum, created_by
entity_revisions	Per-entity checkpoint payload	entity_type, entity_id, operation, snapshot_json
change_events	Mutation ledger	origin, actor_id, operation_id, json_patch, inverse_patch, occurred_at
restore_operations	Restore audit	source_version_id, result_version_id, scope_json, reason
tasks / calendar_events	Work and time	garment_id, status, due_at, assignee_id, event_type, starts_at
ai_jobs	Machine work request	job_type, status, model, prompt_version, input_refs_json, started_at
ai_artifacts	Reviewable candidate	artifact_type, candidate_json, provenance_json, confidence_json, decision
sync_tombstones	Distributed deletion	entity_type, client_id, deleted_at, updated_at

SCHEMA COMPLETENESS

Pages 20-28 diagram every proposed table. Pages 29-31 define responsibility and critical fields. Detailed SQL types, constraints, indexes, and RLS policies are implementation deliverables in work package 2, not assumptions hidden in UI code.

PRODUCT BIBLE

Security, RLS, storage, and publication boundaries

The studio graph is private; public experience is a separate projection.

LAYER	RULE
Authentication	Supabase Auth session is required for all private routes. Public portfolio routes never reuse signed-in in-memory studio state.
Studio membership	RLS checks active studio_members rows; owner, editor, reviewer, and viewer roles are reserved for 3.0 but schema-ready.
Owner data	All private tables enforce studio membership for select and role-appropriate write operations.
Storage	studio-assets bucket is private under studios/{studio_id}/...; portfolio-assets is public but only contains copied publication derivatives.
Publications	Anonymous select is allowed only on publication tables or an RPC/view that returns current public snapshots.
Technical sensitivity	Costs, supplier identities, pattern files, full technical data, fit notes, and raw AI inputs are never public by default.
Downloads	Time-limited signed URLs for private files; checksum and source version recorded for export artifacts.
Audit	Publication, unpublication, restore, role changes, and AI acceptance create immutable change events.

RLS test matrix

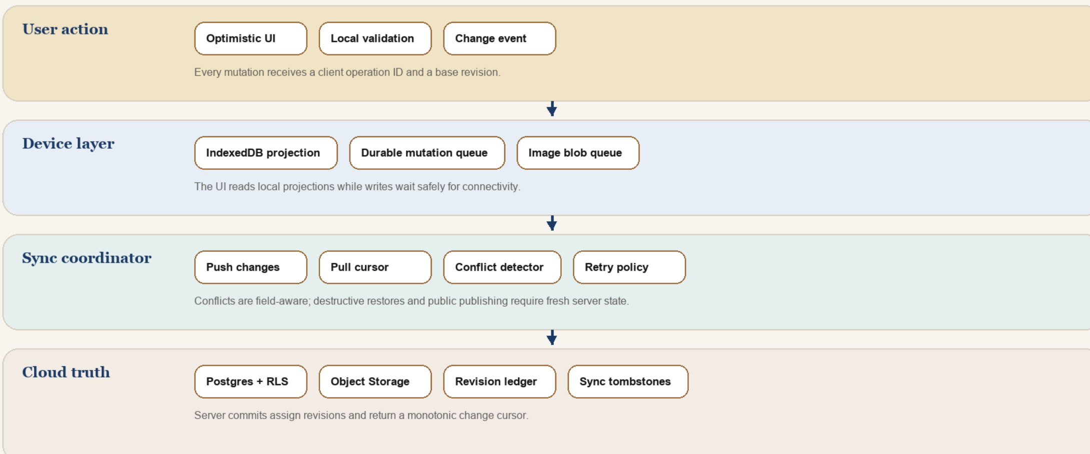
- Anonymous cannot select any private garment, task, note, technical, production, or media row.
- Member of Studio A cannot infer or mutate IDs from Studio B, including storage paths.
- Public reader sees only is_public/current publication snapshots and public derivatives.
- Unpublishing removes anonymous access without deleting private source records or publication history.

PRODUCT BIBLE

Offline-first sync and conflict policy

Preserve the current resilience, but give every domain the same durable contract.

OFFLINE-FIRST DATA FLOW

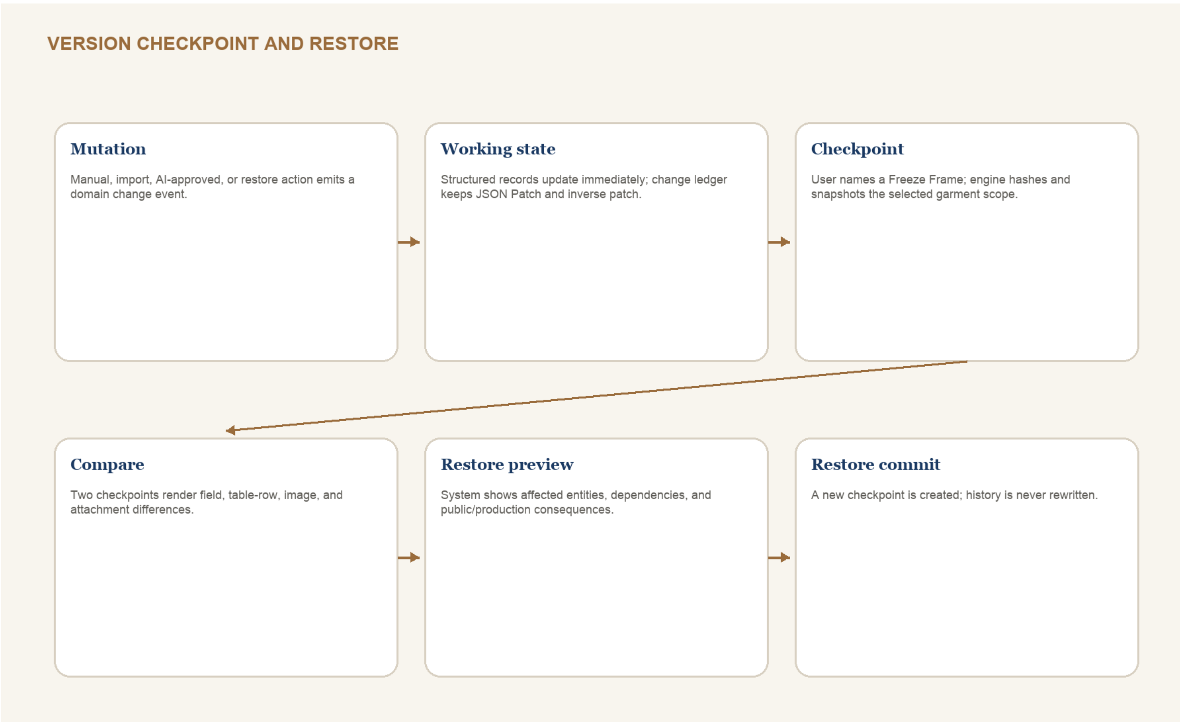


CONFLICT	POLICY
Different fields	Merge automatically; emit one combined change event with both operation IDs.
Same scalar field	Show before, local, and remote values; user selects or writes a new value.
Ordered children	Merge by stable child ID and fractional sort key; warn on same-item moves.
Media	Deduplicate by checksum; preserve separate framing and role relationships.
Delete versus edit	Tombstone wins provisionally; offer restore-as-new-revision when user has unsynced work.
Release / publish / restore	Require fresh server revision and online commit; never queue blindly.

PRODUCT BIBLE

Versioning architecture: Freeze Frames

Git-style confidence without exposing designers to repository mechanics.



2.x version model

- Working state is continuously saved; it is not itself a named version.
- A Freeze Frame is a named checkpoint over a garment scope: all, design, technical, production, editorial, or portfolio.
- Release checkpoints are system-labeled and cannot be deleted while referenced by a production order, export, or publication.
- Long-lived branches are deferred to 3.0. In 2.x, experimentation uses duplicate garment or named draft checkpoint plus restore.
- Restore creates a new checkpoint whose parent is the current version; it never rewrites the timeline.

PRODUCT BIBLE

Version compare, release gates, and restore rules

Comparison must understand product structure, not just JSON text.

ARTIFACT	COMPARE REPRESENTATION	RESTORE BEHAVIOR
Brief and garment metadata	Field before/after with author and time	Selected fields or full design scope
POM / measurements	Row identity by POM point and size; tolerance and actual deltas	Restore rows while preserving later sample evidence unless explicitly selected
BOM	Added/removed/replaced item, quantity, placement, supplier and cost impact	Warn about inventory reservations and open production orders
Construction	Step reorder, operation text, machine/stitch, image callouts	Restore selected sections or complete construction scope
Media / flats	Asset checksum, role, framing, annotation, approval	Never delete source asset if referenced elsewhere
Editorial	Scene/block order, content, theme, live-data source and staleness	Published snapshot stays unchanged until republished
Portfolio	Visibility, case-study copy, order, selected assets	Public cut stays unchanged until explicit publish

RELEASE GATE

A release gate is a named validation ruleset plus a Freeze Frame. Waivers require reason, actor, time, affected rule, and follow-up task. Critical privacy rules cannot be waived.

PRODUCT BIBLE

Design language: The Modern Atelier

Dark, tactile, editorial - with the clarity and discipline of a technical workbench.

TOKEN	VALUE	ROLE
Midnight	#0A0A0A	App background and high-focus canvas
Celestial	#1B3A63	Structural navigation, technical information, calm focus
Golden Ember	#C89B3C	Primary action, approved status, active lens, highlights
Nebula Teal	#2D5C6B	Informational state, material and data relationships
Dusk Bronze	#9A6C3C	Borders, secondary emphasis, tactile detail
Stardust Ivory	#EDE3CF	Primary text on dark surfaces
Deep Espresso	#3D2B1F	Warm supporting surfaces and editorial mood

Typography and shape

- Display: Cinzel Decorative only for page-level or campaign titles. Never use it in dense tables, forms, or long copy.
- Workbench: Inter or system sans. Use tabular numerals for measurements, costs, quantities, and version numbers.
- Radius: 12 for fields, 16 for cards, 24 for presentation surfaces. Technical grids use restrained radius and stronger alignment.
- Borders carry hierarchy; shadows are subtle. Selected, focused, warning, error, and approved states must not rely on color alone.
- Motion is quiet: 160-240 ms for controls, 320-460 ms for editorial transitions; honor reduced motion.

CONTRAST CORRECTION

The current brand can become too dim in long work sessions. Workbench surfaces use lighter text, clearer field boundaries, and stronger focus rings than editorial viewing surfaces.

PRODUCT BIBLE

Interaction, responsive, and accessibility principles

The interface changes density, not capability or vocabulary.

PRINCIPLE	CONTRACT
One primary action	Each screen has one dominant commit or progression action; secondary actions remain available but visually quieter.
Autosave with evidence	Show saved locally, syncing, synced, conflict, and failed. Releases and publications remain explicit commits.
Keyboard workbench	Command palette, table row navigation, add row, duplicate, compare, save checkpoint, and validation shortcuts.
Responsive lenses	Desktop: rail + canvas + Threadline. Tablet: rail collapses, inspector becomes sheet. Mobile: capture and next-move views.
Touch-safe	Minimum 44 px controls; drag has menu alternatives; tables offer row detail cards on narrow screens.
Semantic structure	Real headings, labels, table headers, landmarks, live regions, and error summaries.
Focus and motion	Visible 2 px focus treatment; logical order; focus returns after modal; reduced motion disables nonessential transitions.
Data legibility	Do not encode status only by hue; units live in headers and inputs; decimals and dates are locale-aware.

Accessibility target

Target WCAG 2.2 AA for product UI and public portfolio. Screenshot review is insufficient: automated checks, keyboard walkthroughs, zoom/reflow tests, and screen-reader verification are release requirements.

PRODUCT BIBLE

Component library I: foundations

Tokens and primitives shared by every domain.

COMPONENT	VARIANTS / STATES	REQUIRED BEHAVIOR
AppShell	desktop, tablet, mobile	Primary rail, garment lens bar, Threadline slot, main landmark, skip link
PageHeader	editorial, workbench, field	Title, context, status, one primary action, responsive collapse
Button / IconButton	primary, secondary, quiet, danger	loading, disabled reason, tooltip for icon-only, 44 px touch target
Field	text, number, unit, date, select, combobox	label, hint, error, dirty state, keyboard and autocomplete policy
Card / Surface	standard, selected, warning, presentation	consistent padding, semantic region where useful, no nested-card clutter
Tabs / LensSwitcher	route, local panel, compact	URL for durable lenses; aria-current or tab semantics; overflow handling
Badge / Status	neutral, info, warning, error, approved	icon + label when meaning is critical; never color-only
Dialog / Sheet	confirm, editor, inspector, mobile sheet	focus trap, escape, destructive confirmation, return focus
Toast / SyncStatus	saving, synced, queued, conflict, failed	live region, persistent path to details, no success spam

MIGRATION RULE

Reuse current shared components where semantics and accessibility are sound. Introduce Storybook-style examples and tests before replacing visual treatment.

PRODUCT BIBLE

Component library II: technical and data authoring

Dense tools need predictable selection, units, validation, and batch actions.

COMPONENT	USE	CONTRACT
DataGrid	POM, BOM, cost, QC, grade rules	Sticky header, row identity, keyboard cells, unit-aware columns, row detail fallback
EditableCell	Inline scalar changes	Enter to edit, Escape to cancel, dirty indicator, validation, conflict marker
MeasurementInput	Targets, tolerance, actuals	Explicit unit, decimal precision, range validation, paste support
POMCanvas	Diagram and point anchors	Zoom/pan, keyboard point list, selected anchor sync, no inaccessible canvas-only data
FlatCanvas	Technical view and callouts	Source/revision, annotation anchors, compare overlay, approval status
SequenceBuilder	Construction steps and editorial scenes	Drag plus move menu, stable IDs, bulk duplicate, dependency warnings
RelationshipPicker	Material, component, supplier, asset links	Search existing first, create inline second, show downstream use
InspectorPanel	Selected row/block/asset	Edits selected entity only; dirty-state warning on selection change
ValidationSummary	Release and publish gates	Group by severity and domain; jump to exact field; waiver workflow
DiffViewer	Version comparison	Field, row, order, asset, and prose diff modes; restore selection

Table editing rule

The grid is the fastest authoring surface, not the only readable representation. On mobile, rows become labeled detail cards and batch work is deferred to tablet or desktop.

PRODUCT BIBLE

Component library III: AI, versions, and publishing

High-consequence actions share a visible proposal-preview-commit pattern.

PATTERN	STAGES	NON-NEGOTIABLE DETAILS
AICandidatePanel	queued, running, candidate, accepted, rejected	Input sources, prompt/ruleset version, model, confidence, changed entities, decision
FreezeFrameDialog	scope, label, notes, validation, commit	Shows included domains and unresolved warnings; release labels protected
RestorePreview	source, current, affected, dependency warnings, commit	Creates new version; never silently rolls back public or production artifacts
PublicCutPreview	selection, privacy scan, anonymous preview, publish	Exact public payload, selected assets, source version, staleness
ExportPanel	format, sections, validation, generation, artifact	Template version, source garment version, checksum, deterministic file name
ConflictResolver	base, local, remote, resolution	Field-aware options; preserve both evidence and the final chosen value
ReleaseGate	checks, warnings, waivers, checkpoint, release	Ruleset version and actor recorded; critical privacy failures block commit

SHARED MENTAL MODEL

AI acceptance, restore, release, export, and publish all work the same way: inspect inputs, preview consequences, resolve blockers, commit an auditable artifact.

PRODUCT BIBLE

Complete screen inventory

Every screen has one primary owner and one primary user action.

AREA	SCREENS	PRIMARY ACTION
Access	Sign in, sign up, recovery, first-studio setup	Enter or establish private studio
Today	Dashboard, command palette, activity/inbox	Choose or create the next move
Garments	Library, collection workspace, garment overview, Design Studio, moodboard, media library	Define and navigate garment intent
Technical	Technical home, flats, POM map, measurement set, grading, BOM, construction, technical files, export	Create and release structured specification
Production	Production home, suppliers, factories, sample rounds, fit sessions, costing, production order, QC, timeline	Approve evidence and control risk
Libraries	Material Vault, material detail, Component Library, component detail, supplier detail, templates	Reuse verified studio knowledge
Editorial	Editorial Library, collection setup, scene builder, viewer, export	Compose and deliver a garment story
Portfolio	Manager, profile editor, project settings, Public Cut preview, publish history	Curate and publish safe public work
Public	Portfolio home, project case study, editorial viewer, technical excerpt/download	Understand the designer and work
Platform	Versions, diff, restore, tasks, calendar, AI jobs, settings, sync/backup	Control change, work, policy, and resilience

WIREFRAME COVERAGE

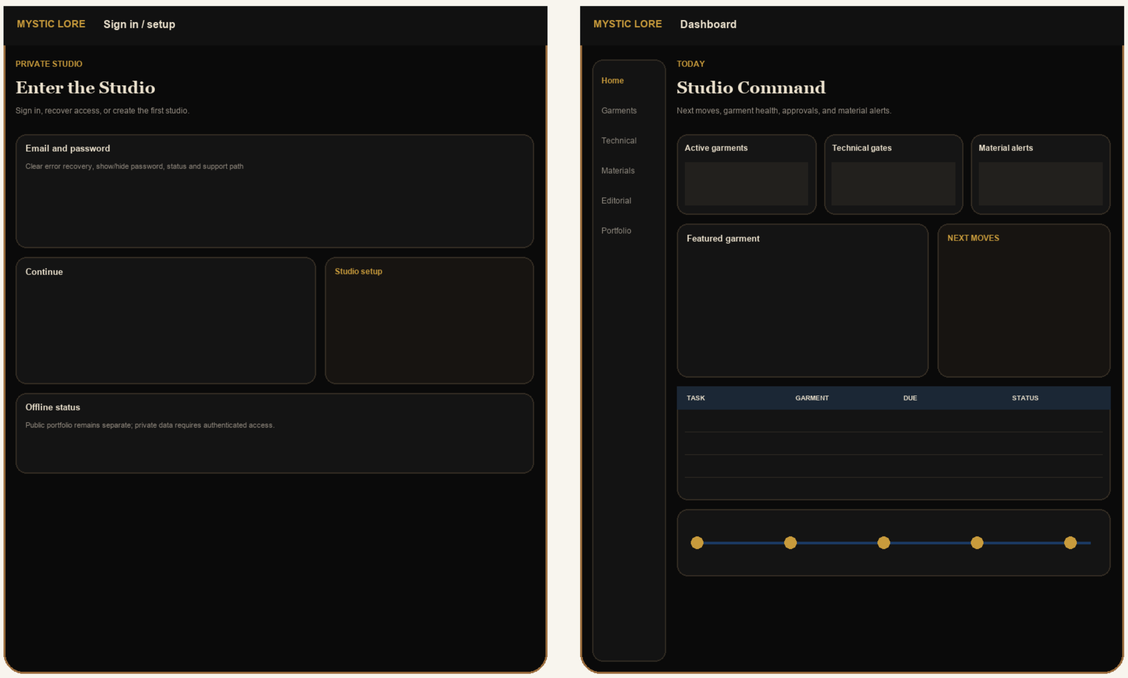
Pages 42-56 cover every screen above. Closely related detail/setup screens use the same shell shown for their area and are represented by a named tab, inspector, or modal within the grouped wireframe.

WIREFRAME 1 / 15

Access and daily cockpit

Sign in, sign up/setup pattern, Dashboard.

ACCESS AND DAILY COCKPIT



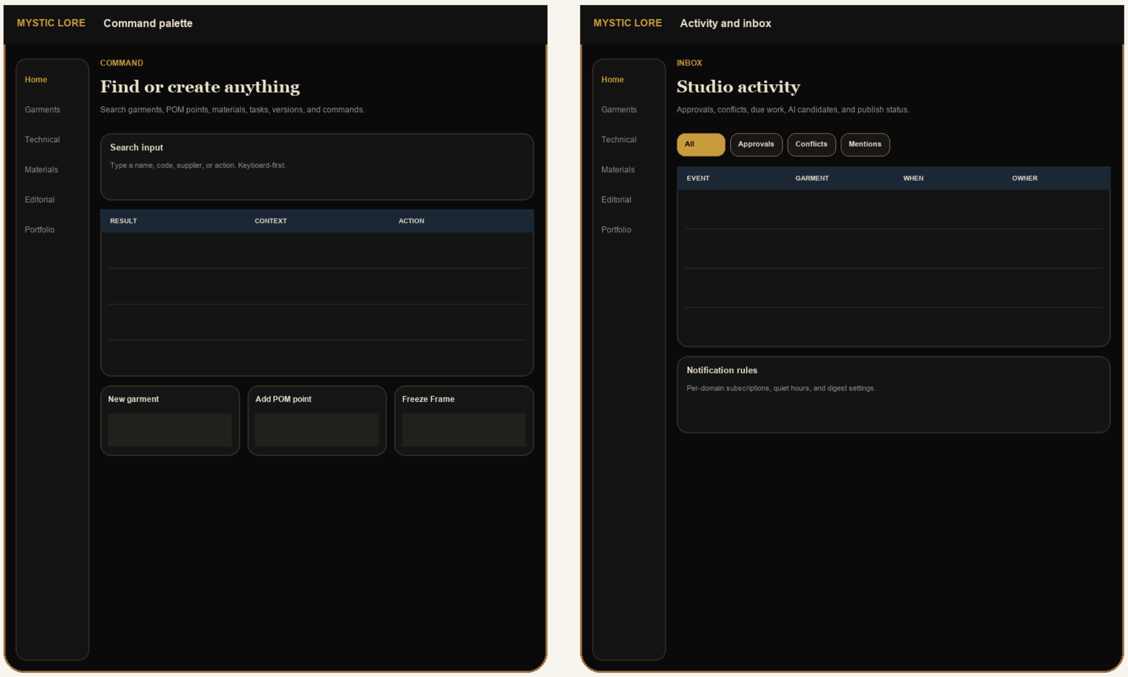
Low-fidelity interaction contract. Visual design is governed by pages 36-40; labels and information hierarchy are intentional.

WIREFRAME 2 / 15

Global command and activity

Command palette, search/create actions, activity inbox and notification rules.

GLOBAL UTILITY SURFACES



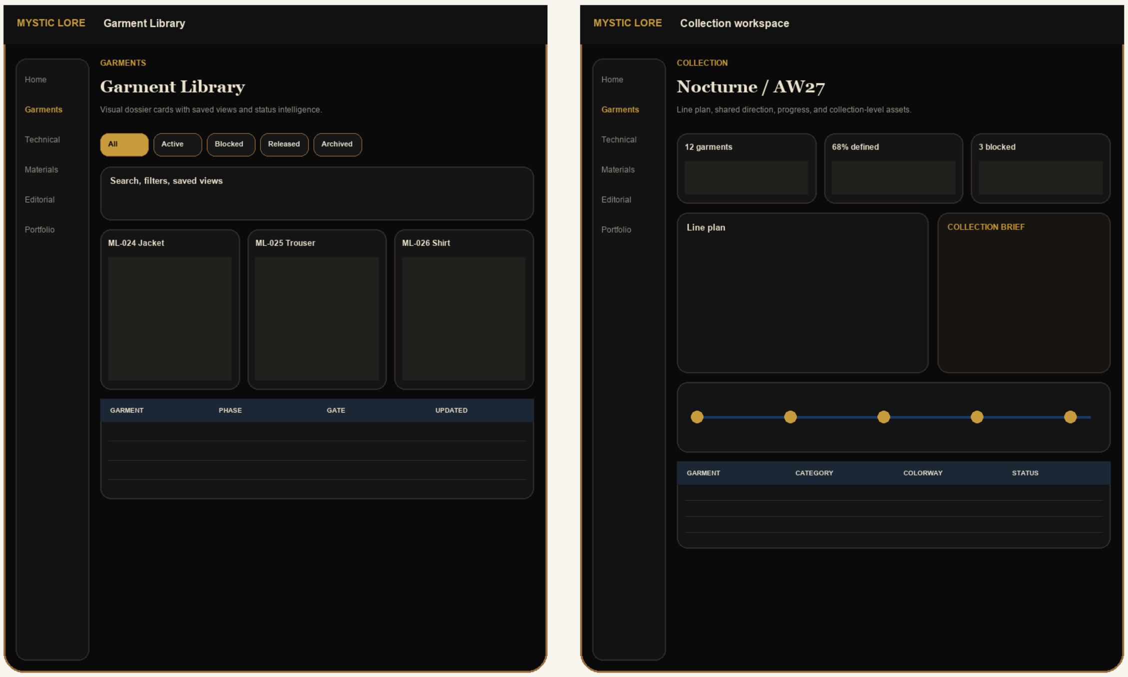
Low-fidelity interaction contract. Visual design is governed by pages 36-40; labels and information hierarchy are intentional.

WIREFRAME 3 / 15

Garment Library and collections

Garment search, saved views, line plan, and collection brief.

GARMENT ORGANIZATION



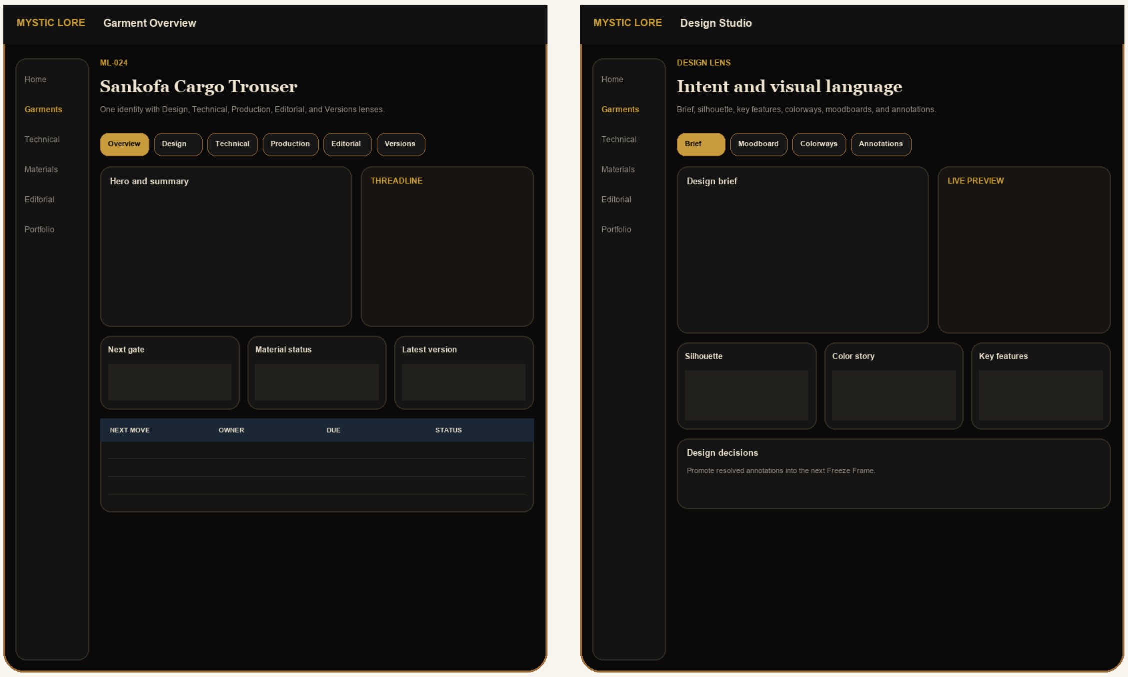
Low-fidelity interaction contract. Visual design is governed by pages 36-40; labels and information hierarchy are intentional.

WIREFRAME 4 / 15

Garment overview and Design Studio

Six lenses, Threadline rail, brief, colorways, and design decisions.

GARMENT WORKSPACE LENSES



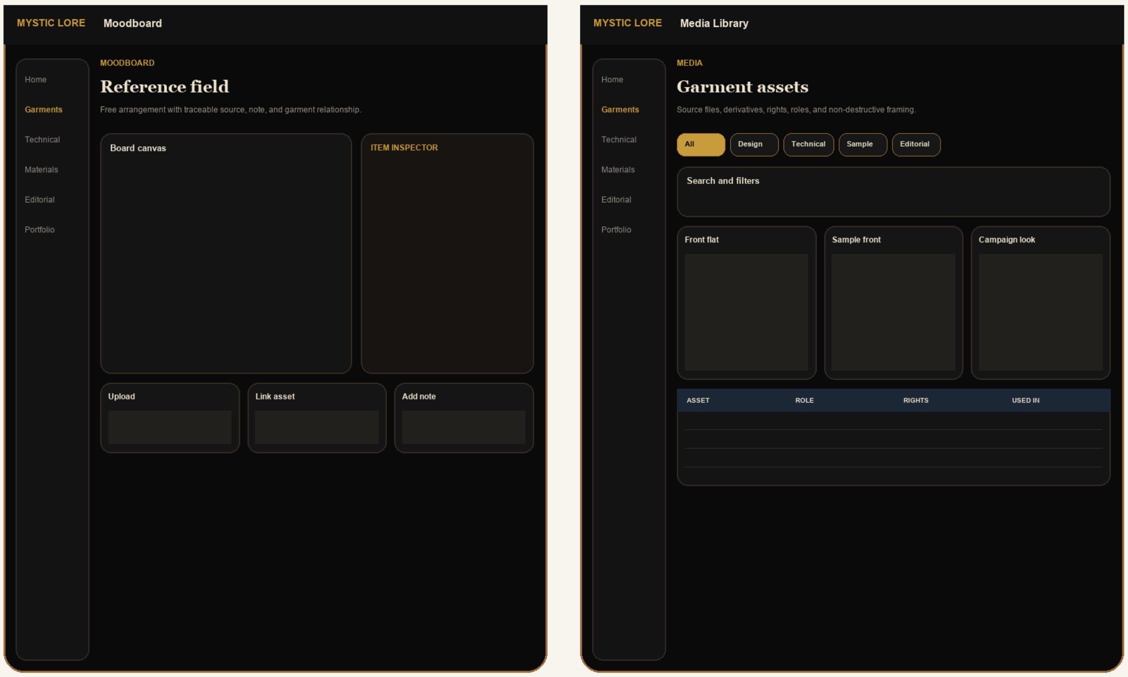
Low-fidelity interaction contract. Visual design is governed by pages 36-40; labels and information hierarchy are intentional.

WIREFRAME 5 / 15

Moodboards and media

Reference canvas, asset rights, roles, derivatives, and annotations.

VISUAL DEVELOPMENT



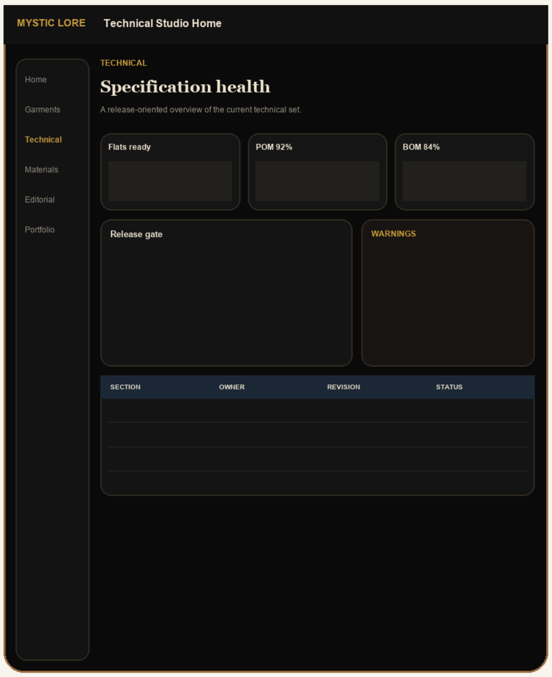
Low-fidelity interaction contract. Visual design is governed by pages 36-40; labels and information hierarchy are intentional.

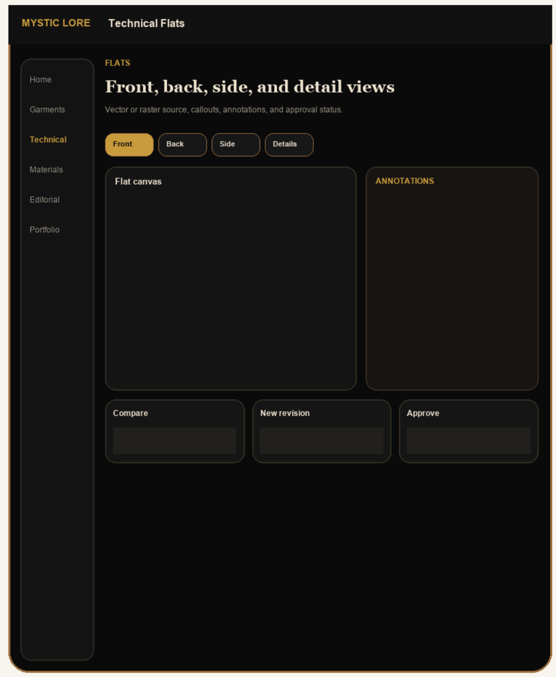
WIREFRAME 6 / 15

Technical Studio and flats

Release health, warnings, flat views, annotations, compare, and approval.

TECHNICAL STUDIO ENTRY





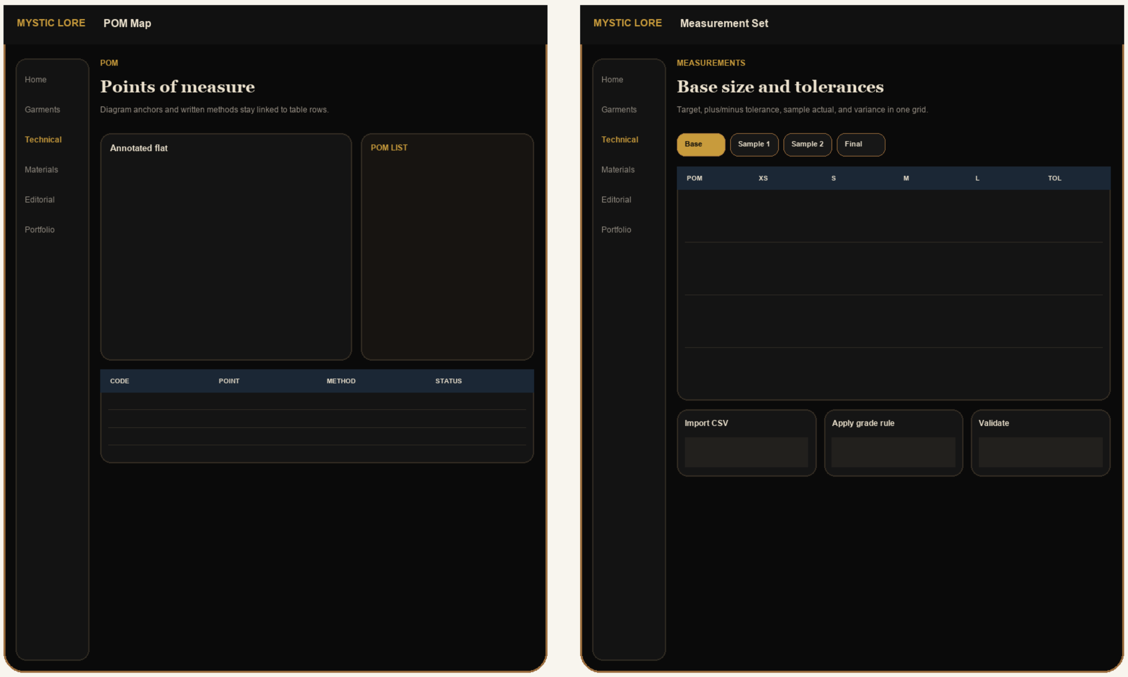
Low-fidelity interaction contract. Visual design is governed by pages 36-40; labels and information hierarchy are intentional.

WIREFRAME 7 / 15

POM and measurement sets

Diagram anchors, methods, base targets, tolerances, sample actuals, and validation.

POM AND MEASUREMENT AUTHORING



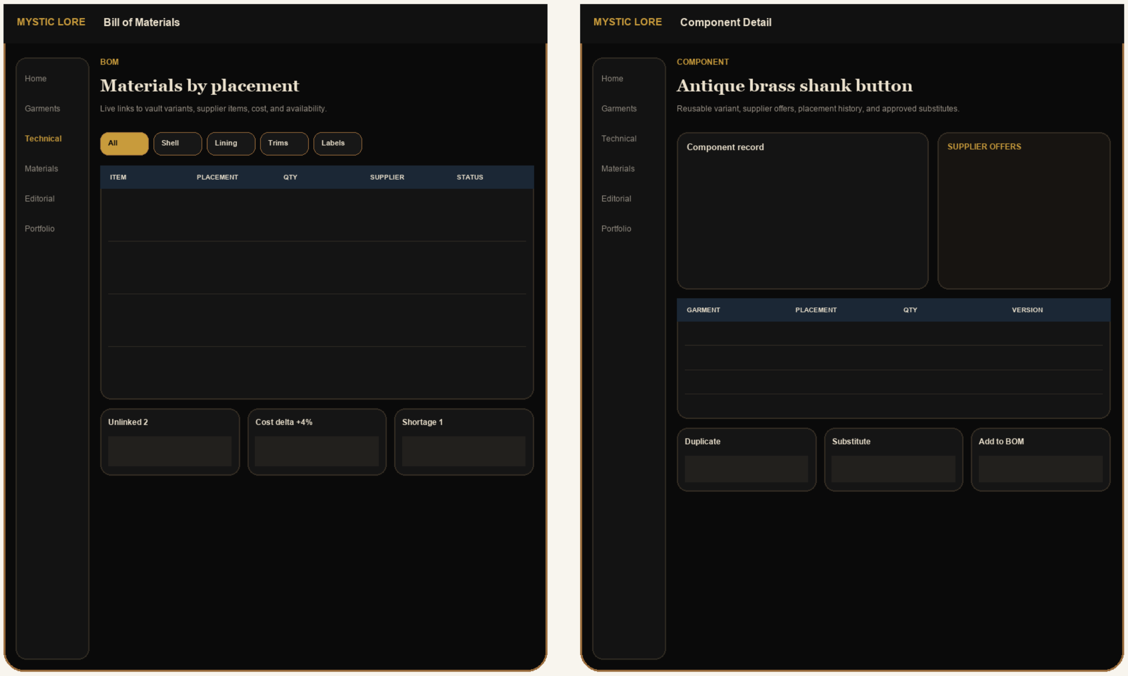
Low-fidelity interaction contract. Visual design is governed by pages 36-40; labels and information hierarchy are intentional.

WIREFRAME 8 / 15

BOM and components

Variant links, placements, quantities, supplier offers, substitutes, and cost impact.

BOM AND COMPONENT RELATIONSHIPS



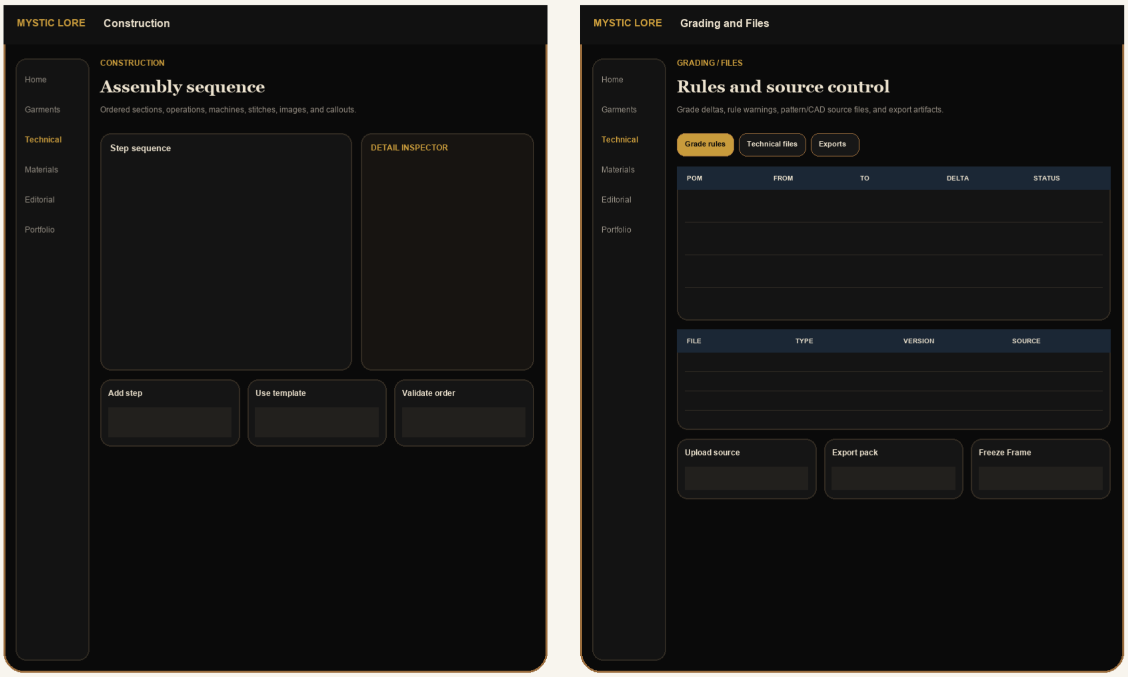
Low-fidelity interaction contract. Visual design is governed by pages 36-40; labels and information hierarchy are intentional.

WIREFRAME 9 / 15

Construction, grading, and files

Assembly sequence, detail inspector, grade rules, source files, and exports.

CONSTRUCTION, GRADING, AND FILES



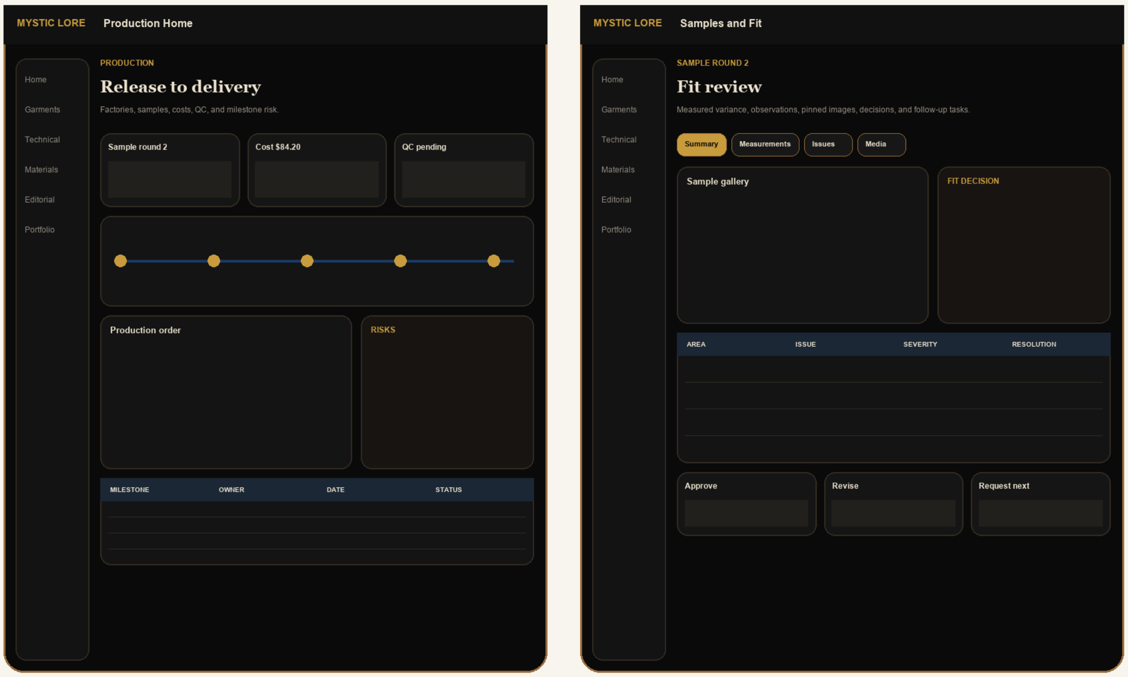
Low-fidelity interaction contract. Visual design is governed by pages 36-40; labels and information hierarchy are intentional.

WIREFRAME 10 / 15

Production, samples, and fit

Production overview, sample rounds, fit measurements, issues, and decisions.

PRODUCTION AND SAMPLING



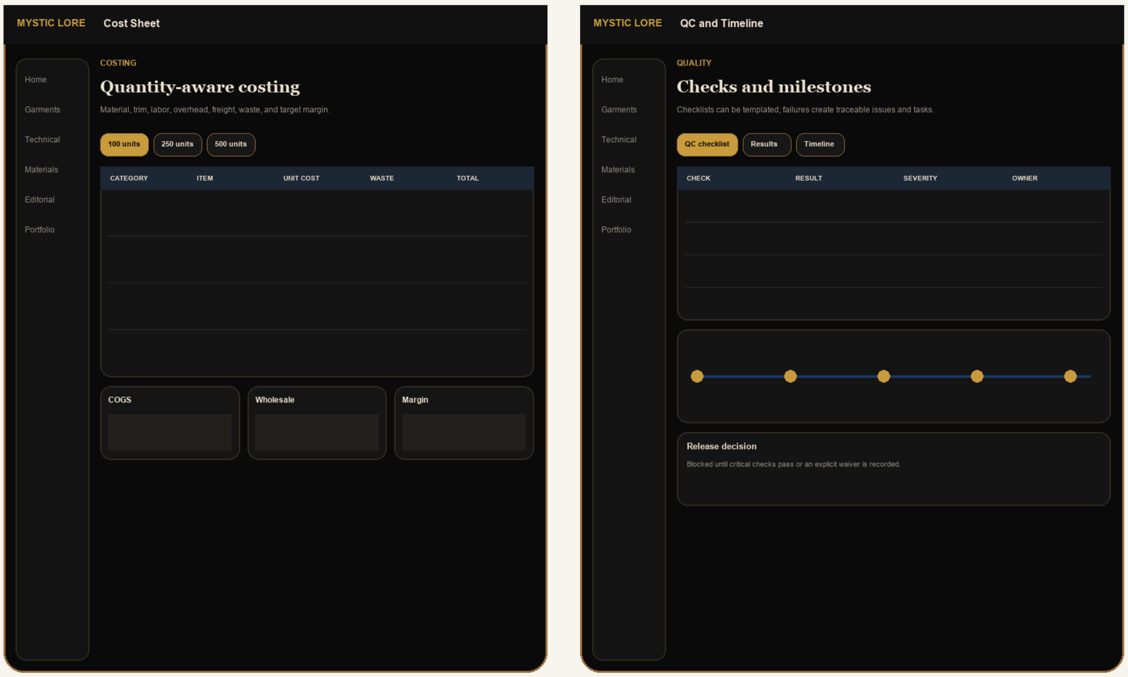
Low-fidelity interaction contract. Visual design is governed by pages 36-40; labels and information hierarchy are intentional.

WIREFRAME 11 / 15

Costing, QC, and timeline

Quantity scenarios, quality checks, waivers, release decision, and milestones.

COSTING, QC, AND TIMELINE



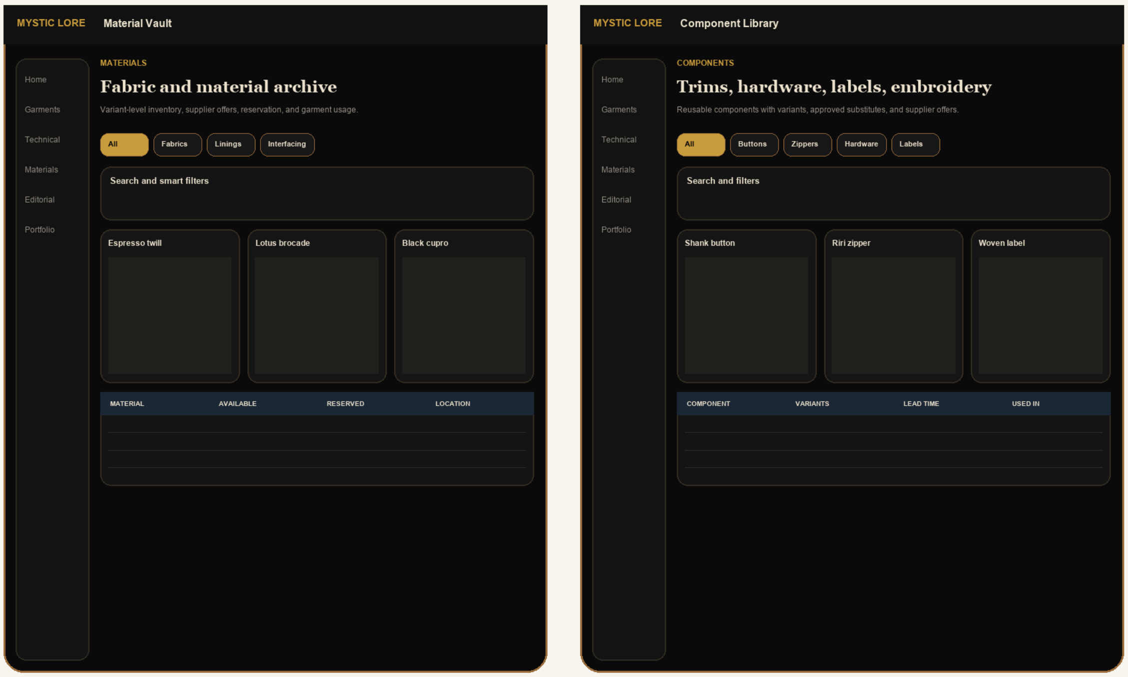
Low-fidelity interaction contract. Visual design is governed by pages 36-40; labels and information hierarchy are intentional.

WIREFRAME 12 / 15

Material and Component Libraries

Searchable reusable records, inventory, supplier offers, variants, and usage.

REUSABLE STUDIO LIBRARIES



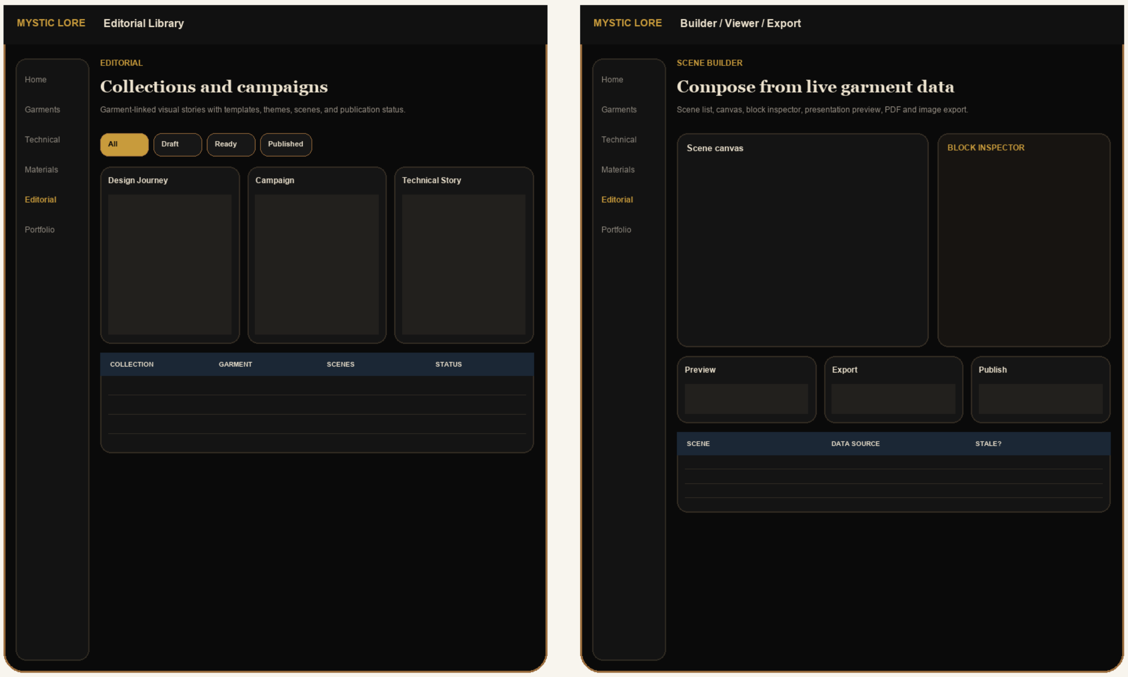
Low-fidelity interaction contract. Visual design is governed by pages 36-40; labels and information hierarchy are intentional.

WIREFRAME 13 / 15

Editorial Collections

Library, setup, scene builder, live-data blocks, viewer, PDF/image export, and publish.

EDITORIAL COLLECTION SYSTEM



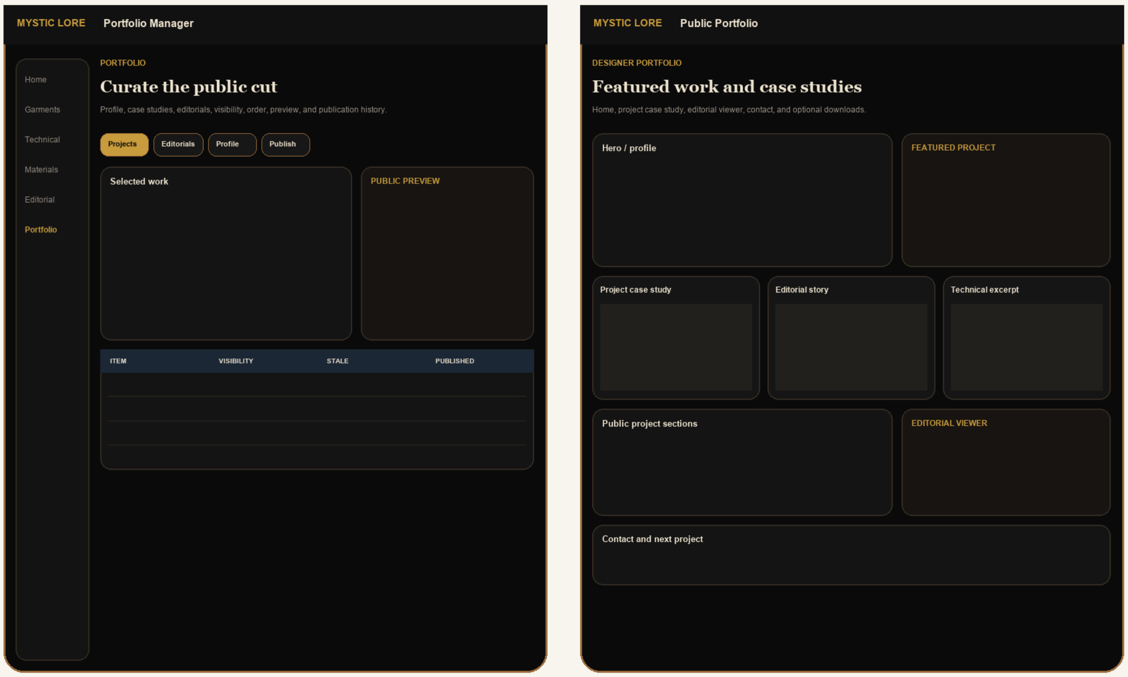
Low-fidelity interaction contract. Visual design is governed by pages 36-40; labels and information hierarchy are intentional.

WIREFRAME 14 / 15

Portfolio manager and public experience

Profile, project/editorial selection, Public Cut preview, public home, case study, and editorial viewer.

PORTFOLIO MANAGEMENT AND PUBLIC EXPERIENCE



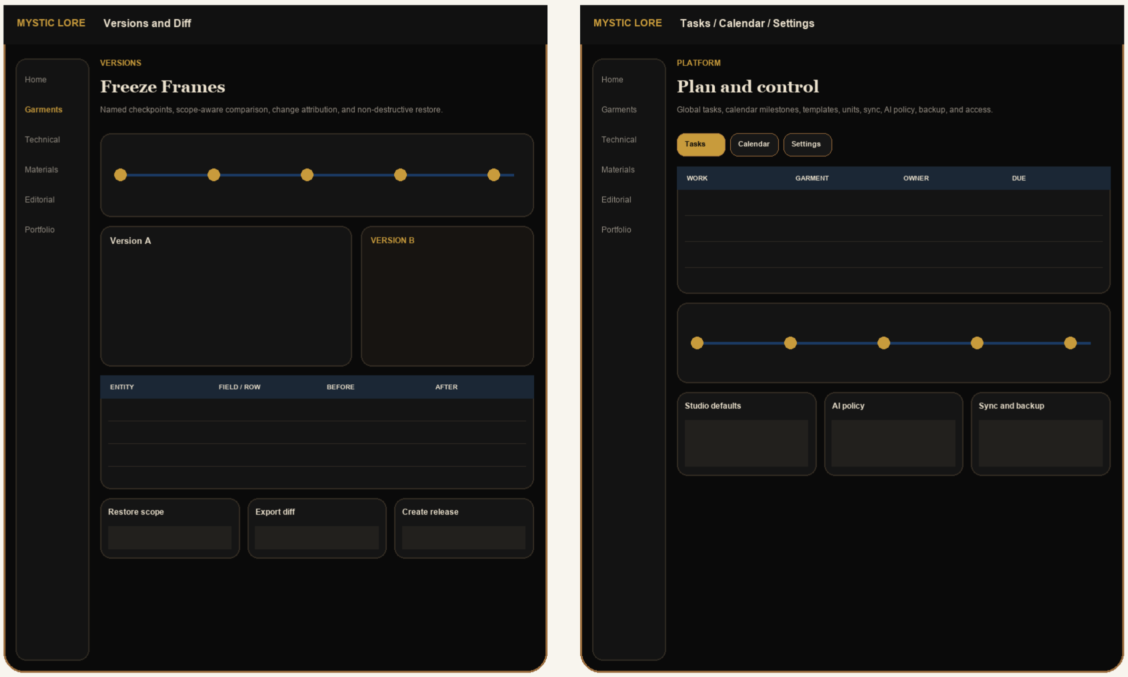
Low-fidelity interaction contract. Visual design is governed by pages 36-40; labels and information hierarchy are intentional.

WIREFRAME 15 / 15

Versions, tasks, calendar, and settings

Freeze Frames, structural diff, restore, planning, AI policy, sync, and backup.

HISTORY, PLANNING, AND SYSTEM CONTROLS



Low-fidelity interaction contract. Visual design is governed by pages 36-40; labels and information hierarchy are intentional.

PRODUCT BIBLE

AI workflows and trust model

AI accelerates structured work while the designer remains the committing authority.

WORKFLOW	INPUTS	CANDIDATE OUTPUT	COMMIT GATE
Technical flat generation	Approved reference photos, view request, garment brief	Front/back/detail flat asset plus source mapping	Visual fidelity review and flat revision approval
POM assistance	Garment type, flats, studio POM template	Suggested point list, methods, anchors, tolerances	Technical designer reviews each point and unit
BOM assistance	Design brief, materials, components, similar approved garments	Candidate BOM rows with relationship suggestions	Resolve every item link, quantity, unit, and placement
Construction recommendations	Flats, BOM, studio templates	Ordered candidate steps and callouts	Human confirms operation, machine, stitch, seam and order
Tech pack validation	Structured technical set and ruleset	Errors, warnings, conflicts, and jump links	Critical failures block release; waiver audit for allowed warnings
Editorial generation	Selected garment facts, approved images, tone and template	Scene outline and editable live-data blocks	Preview, source staleness check, human copy and visual review
Portfolio automation	Approved case-study facts, editorial, public-safe selection	Draft case study and Public Cut selection	Privacy scan plus explicit anonymous preview and publish

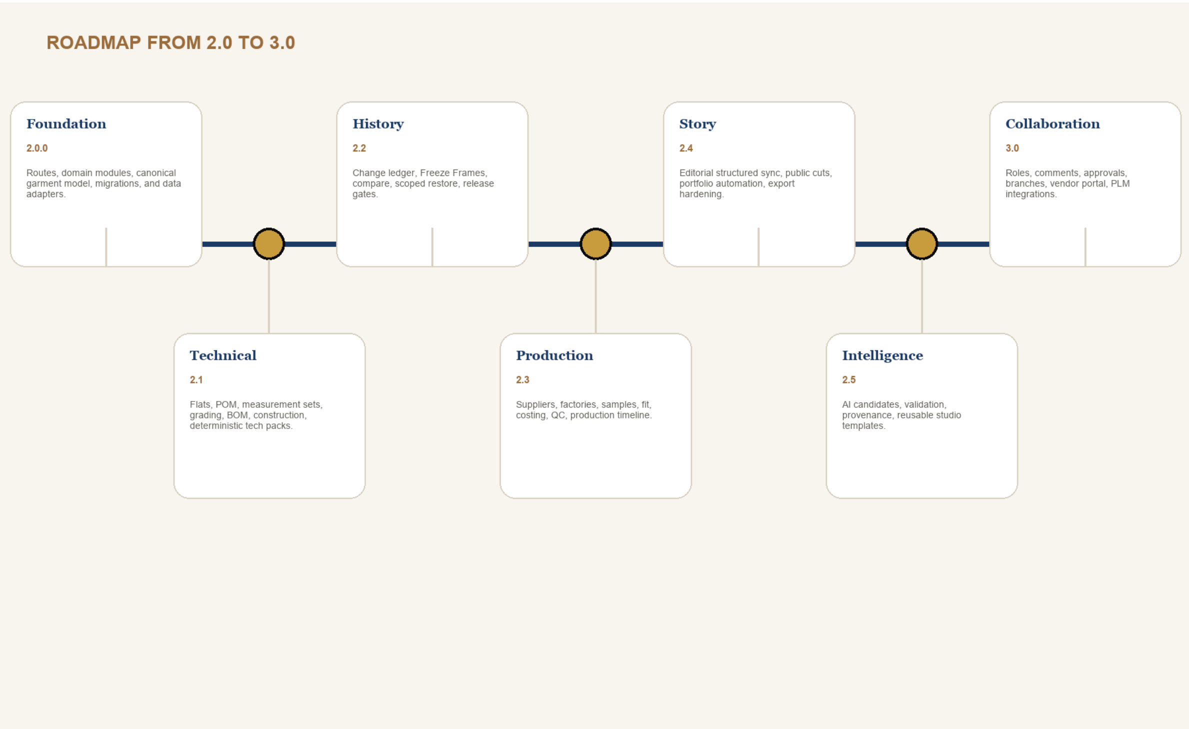
Trust contract

- Every job records model, prompt template version, structured inputs, source entity versions, generated artifacts, and user decision.
- Candidates never write production, technical release, public, cost, supplier, or measurement records directly.
- Accepted candidates emit the same domain commands and change events as manual edits.
- The UI labels generated, accepted, modified-after-generation, and rejected states. Confidence is contextual, never a truth score.

PRODUCT BIBLE

Future roadmap: 2.0 to 3.0

Ship coherence first, technical depth second, collaboration last.



RELEASE	DEFINITION OF SUCCESS
2.0 Foundation	Existing core features run on route modules and the canonical garment model without data loss.
2.1 Technical	Designer can release a validated, reproducible tech pack from structured data.
2.2 History	High-value changes are comparable and restorations are safe and auditable.
2.3 Production	Sample, fit, cost, order, milestone, and QC decisions reference the correct version.
2.4 Story	Editorials sync as structured data; public cuts are exact, previewable, and stale-aware.
2.5 Intelligence	AI candidates reduce setup time without weakening provenance, privacy, or quality gates.
3.0 Collaboration	Roles, approvals, comments, branches, vendor portal, and integrations build on proven domain boundaries.

PRODUCT BIBLE

Codex implementation order

Each work package has a bounded target, explicit dependencies, and a safe stopping point.

WP	BUILD TARGET	EXIT CONDITION
0	Characterization tests, screenshots, data export fixture, architecture decision records	Current behavior and migration input are reproducible
1	Route modules, garment workspace shell, domain folders, typed command/repository interfaces	No feature change; build and route parity pass
2	Proposed schema foundation, RLS, storage paths, read-through adapters, migration dry run	Legacy fixture migrates and round-trips with zero loss
3	Garment, collection, design, media, Material Vault, and component domains	Core relationships replace duplicated project/material fields
4	Technical spec, flats, POM, measurements, grading, BOM, construction, validation, export	One seeded garment produces a reproducible approved tech pack
5	Change events, Freeze Frames, structural diff, scoped restore, release gates	Restore creates a new version and preserves linked evidence
6	Suppliers, factories, samples, fit, costing, orders, QC, timeline	Production decisions pin to released garment version
7	Editorial normalization and migration from lookbook/editorial legacy models	Collections sync cross-device and exports match approved sources
8	Portfolio curation, Public Cut preview, publication schema, staleness, public routes	Anonymous access exposes only selected snapshot data
9	AI jobs/artifacts, candidate panels, provenance, acceptance commands	AI cannot bypass domain validation or write private/public records directly
10	Performance, accessibility, responsive field mode, observability, beta migration	Page 60 gates pass on representative data and devices

CODEX PROMPT TEMPLATE

Implement WP N using Product Bible pages X-Y. Preserve unrelated user changes. Start with characterization tests, name migrations, keep public and private schemas separate, and stop when the listed exit condition is verified.

PRODUCT BIBLE

Quality gates, open decisions, and glossary

The redesign is ready to code only when the work package can pass these checks.

GATE	DEFINITION OF DONE
Data	Migration is reversible or recoverable, idempotent where appropriate, typed, indexed, and verified against a representative export.
Privacy	RLS tests prove cross-studio isolation and anonymous denial; public payload is inspected for excluded fields.
Versioning	Mutation origin and actor recorded; release/export/public artifacts reference source version; restore is non-destructive.
UX	Primary path, empty/loading/error/conflict states, keyboard path, mobile fallback, and destructive confirmation are implemented.
Accessibility	WCAG 2.2 AA target checked with automated tests plus keyboard, zoom/reflow, and screen-reader walkthrough.
Reliability	Offline queue, retry, conflict, tombstone, image upload, and reload behavior verified.
Performance	Large garment records, 1,000-row technical grids, image-heavy editorials, and public pages remain responsive.
Documentation	Schema, command contracts, validation rules, screen states, and change log updated in this bible or linked ADR.

Open product decisions before affected work package

- Unit policy: one studio unit with export conversion, or mixed units per measurement set? Recommendation: one spec unit, explicit converted display.
- Garment-to-editorial cardinality: one primary garment plus optional supporting garments? Recommendation: support one primary plus many supporting.
- 2.x collaboration: invite reviewers before 3.0? Recommendation: public/private review links only; no private studio membership until role UX is complete.

Glossary

Garment Data Engine: canonical relationship and event layer. Freeze Frame: named immutable garment checkpoint. Public Cut: sanitized immutable publication snapshot. Threadline: persistent garment context rail. Release Gate: validation ruleset plus approved Freeze Frame. Live-data block: editorial block linked to structured garment data with staleness tracking.